RUNTIME LEARNING IN OPEN WORLD: ADAPTIVE BLENDING AND COGNITIVE SAMPLING

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ABSTRACT

This paper proposes **OpenRun**, a framework that enables safety-critical autonomous systems to perform runtime learning in open-world environments. Here, *open-world environments* refer to environments whose states, conditions, and operational scenarios cannot be fully known or represented prior to deployment and may evolve or newly emerge during operation. **OpenRun** comprises a Learning-Student, a Safety-Teacher, and an Action-Blender. The Learning-Student builds on actor-critic reinforcement learning and introduces a novel cognitive batch sampling strategy that prioritizes experiences based on temporal recency, occurrence frequency, and contextual similarity, significantly enhancing learning efficiency. The Safety-Teacher operates in parallel with the Learning-Student, providing verifiable safety protocol that enables safe learning for a broad range of action policies, from planning and control to vision-to-action policies. The Action-Blender introduces a novel safety- and exploration-aware blending mechanism that adaptively blends the actions of the Learning-Student and Safety-Teacher to enhance safety assurance while preserving exploration. We further establish theoretical foundations demonstrating how these innovations collectively enable safer and more efficient runtime learning in open-world environments. Extensive experiments with a quadruped robot and an off-road autonomous vehicle in challenging real-world and simulated wild forests validate the effectiveness and distinctive capabilities of **OpenRun**, as well as its generalizability across functions, platforms, and environments.

1 INTRODUCTION

AI is increasingly being integrated into autonomous systems, offering the potential to transform industrial operations and drive economic growth through advanced autonomous intelligence. Although significant progress has been made, much of it has focused on structured or human-centered environments, such as factories and warehouses. However, many emerging applications, such as navigation and emergency supply delivery in disaster-stricken areas and remote forests, require autonomous systems to operate in the open world. These open-world environments present the following fundamental challenges to achieving robust and adaptive intelligence.

Challenge 1: Non-Stationary and Complex Operating Environments. Existing embodied-AI systems predominantly follow a train-then-deploy paradigm, in which intelligence is learned offline and subsequently deployed as a fixed capability. This paradigm is vulnerable to domain shifts and sim-to-real gaps, which can degrade performance when systems encounter conditions beyond their training distributions. Considerable progress has been made toward mitigating these gaps through simulation, domain adaptation, and transfer learning (Nagabandi et al., 2018; Cao et al., 2022; Imai et al., 2022; Yang et al., 2022; Tan et al., 2018). However, such gaps remain fundamentally unavoidable in the open-world environments, where operating conditions are inherently complex, non-stationary, and impossible to fully characterize a priori.

Challenge 2: Experience Misalignment in Safety-Critical Systems. Autonomous systems operating in the physical world naturally generate temporally correlated, sequential experiences. However, most existing learning methods rely on random experience sampling (Rana et al., 2021; Li et al.; Cheng et al., 2019b; Sasso et al., 2023), where replayed experiences can be poorly aligned with the temporal and state distributions encountered by the system at runtime. Such experience misalignment can reduce learning efficiency and increase the likelihood of erroneous learning updates and unsafe

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actions. This concern is especially acute in safety-critical autonomous systems, where even a single erroneous action can lead to catastrophic consequences within seconds.

Related Work: Substantial efforts have been devoted to developing safe reinforcement learning (RL) frameworks with the potential to enable runtime learning in the open world. Relevant approaches include two broad categories: i) Runtime-Assurance RL, including neural Simplex (Phan et al., 2020), model predictive shielding (Bastani, 2021; Banerjee et al., 2024), and Real-DRL (Mao et al., 2025); and ii) Residual RL, including fixed residual learning (Johannink et al., 2019a; Cao et al., 2024) and dynamic residual learning (Cao et al., 2026). These frameworks have demonstrated improved learning efficiency and safety in simulated environments (Phan et al., 2020; Bastani, 2021; Banerjee et al., 2024; Johannink et al., 2019a; Cao et al., 2024) and structured indoor settings (Mao et al., 2025; Cao et al., 2026). However, whether these benefits extend to complex open-world environments remains an open question (**Challenge 1**). Moreover, these frameworks primarily focus on learning control policies from low-dimensional system states. While many autonomous missions, such as navigation and search and rescue, are vision-based. Extending these frameworks to learn vision-to-action policies while retaining their learning efficiency and safety assurance presents another open challenge. Finally, their predominant reliance on random batch sampling leaves experience misalignment largely unaddressed (**Challenge 2**).

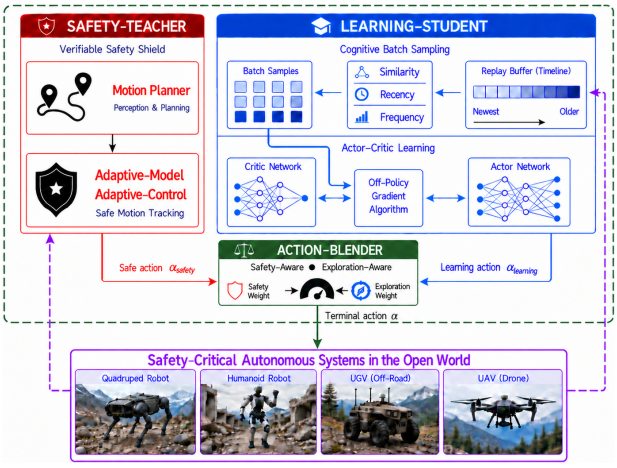


Figure 1: Runtime Learning Framework

Our Approach: To address Challenges 1 and 2, as well as the open problems of existing safe RL frameworks, we propose the **OpenRun** framework, which enables safety-critical autonomous systems to perform high-performance runtime learning in open-world environments. Illustrating in Fig. 1, **OpenRun** consists of three interactive modules: Learning-Student, Safety-Teacher, and Action-Blender, summarized below.

- **Learning-Student**, which builds on actor-critic RL and introduces a novel cognitive batch sampling mechanism inspired by human cognition to prioritize temporally recent, frequently occurring, and contextually similar experiences for learning, significantly improving learning efficiency and accelerating runtime adaptation.
- **Safety-Teacher**, which is a non-learning design dedicated exclusively to safety-critical functions. It generalizes runtime learning across a broad spectrum of functions, ranging from planning and control to end-to-end vision-to-action policies.
- **Action-Blender**, which adaptively blends the real-time actions of the Learning-Student and Safety-Teacher with both safety and exploration awareness for enhancing safety assurance while preserving effective exploration space.

2 PRELIMINARIES

Using the notation summarized in Appendix A, we consider autonomous systems whose dynamics take the following general form:

$$\mathbf{s}(k+1) = \mathbf{f}(\mathbf{s}(k), \mathbf{a}(k)), \quad k \in \mathbb{N}, \tag{1}$$

where $\mathbf{s}(k) \in \mathbb{R}^n$ denotes the real-time system state and $\mathbf{a}(k) \in \mathbb{R}^m$ denotes the real-time action. For safety-critical autonomous systems, the system states are required to always remain within:

$$\text{Safety Set: } \mathbb{S} \triangleq \left\{ \mathbf{s} \in \mathbb{R}^n \middle| \, \mathbf{C} \cdot \mathbf{s} < \mathbf{1}_p, \text{ with } \mathbf{C} \in \mathbb{R}^{p \times n} \text{ and } \mathbf{s} \in \mathbb{R}^n \right\}, \tag{2}$$

where $\mathbf{C}$ specifies $p \in \mathbb{N}$ safety constraints. Based on the above system dynamics and safety set, we formally present the safety definition for safety-critical autonomous systems as follows.

Definition 1 (Safety Guarantee)**.** *Given the safety set $\mathbb{S}$ in Eq. (2), the autonomous system described by Eq. (1) is said to have a* safety guarantee *if, for any current state $\mathbf{s}(k) \in \mathbb{S}$, the system state remains within the safety set at all future times, i.e., $\mathbf{s}(k+\tau) \in \mathbb{S}, \forall \tau \in \mathbb{N}$.*

Remark 1. *The safety set in Eq.* (2) *is defined by linear inequalities. With appropriate state representations, this formulation can equivalently capture a broad class of both linear and nonlinear safety conditions, as illustrated by the concrete examples in Appendix B.*

With these preliminaries in place, we next present the design and theoretical foundations of **OpenRun**.

3 Action-Blender Module

Referring to Fig. 1, the final execution action (i.e., $\mathbf{a}$ in Eq. (1)) can be mathematically described as

$$\mathbf{a} = \beta(\mathbf{s}) \cdot \mathbf{a}_{\mathsf{safety}} + (1 - \beta(\mathbf{s})) \cdot \mathbf{a}_{\mathsf{learning}} \tag{3}$$

which adaptively blends the action $\mathbf{a}_{\mathsf{safety}}$ from the Safety-Teacher with the action $\mathbf{a}_{\mathsf{learning}}$ from the Learning-Student. Here, $\beta(\mathbf{s}) \in (0, 1]$ denotes the adaptive blending weight. The key question here is how to design $\beta(\mathbf{s})$ to enhance safety assurance while preserving learning exploration. To this end, we establish the underlying safety and exploration conditions. Specifically, we first introduce:

$$\text{Exploration Region: } \mathbb{E} \triangleq \left\{ \mathbf{s} \in \mathbb{R}^n | \, \mathbf{s}^\top \cdot \mathbf{P} \cdot \mathbf{s} \leq 1 \right\}, \tag{4}$$

where $\mathbf{P}$ is automatically obtained by solving the following constrained optimization problem (via convex optimization toolbox solvers (Boyd et al., 1994; Grant et al., 2009)), with $\mathbf{C}$ given in Eq. (2).

$$\mathbf{P} = \arg\max_{\widehat{\mathbf{P}} \succ 0} \{\log(\det(\widehat{\mathbf{P}}^{-1}))\}, \text{ subject to } \widehat{\mathbf{P}} \succ 0 \text{ and } \mathbf{C} \cdot \widehat{\mathbf{P}}^{-1} \cdot \mathbf{C}^\top \prec \mathbf{I}_p. \tag{5}$$

Finally, given system dynamics in Eq. (1), we impose the design requirement on Safety-Teacher:

$$\mathbf{f}^\top(\mathbf{s}, \mathbf{a}_{\mathsf{safety}}) \cdot \mathbf{M}(\mathbf{s}) \cdot \mathbf{f}(\mathbf{s}, \mathbf{a}_{\mathsf{safety}}) < \alpha, \text{ and } \bigcup_{\mathbf{s} \in \mathbb{S}} \left\{ \widehat{\mathbf{s}} \in \mathbb{R}^n \,\middle|\, \widehat{\mathbf{s}}^\top \cdot \mathbf{M}(\mathbf{s}) \cdot \widehat{\mathbf{s}} \leq 1 \right\} = \mathbb{S}, \ \forall \mathbf{s} \in \mathbb{S} \tag{6}$$

where $0 < \alpha < 1$ and $\mathbf{M}(\mathbf{s}) \succ 0$. We next derive the theoretical basis for the adaptive weight design.

3.1 Theoretical Basis

In this section, we analyze how the blending weight affects safety assurance and derive a critical lower bound on this weight under the following assumption on the system dynamics.

Assumption 1 (Control-Affine Dynamics)**.** *The dynamics of autonomous systems are affine in the control input. Specifically, the dynamics in Eq.* (1) *can be expressed as:*

$$\mathbf{s}(k+1) = \mathbf{f}(\mathbf{s}(k), \mathbf{a}(k)) = \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}(k) + \mathbf{h}(\mathbf{s}(k)), \tag{7}$$

where $\mathbf{A}(\mathbf{s})$ *and* $\mathbf{B}(\mathbf{s})$ *are known, generally nonlinear matrix-valued functions of the state, and* $\mathbf{h}(\mathbf{s})$ *represents the unknown model mismatch.*

Assumption 1 is broadly applicable in practice for two key reasons. First, many autonomous systems, such as autonomous vehicles (Rajamani, 2011), quadruped robots (Di Carlo et al., 2018), UAVs (Dydek et al., 2012), humanoid robots (Li et al., 2026), and power systems (Mao & Zhang, 2016), admit control-affine representations. Second, the control actions in both model-based and data-driven control are usually state-dependent, allowing unknown model mismatch to be reasonably represented as a state-dependent term. Under this assumption, we next present a theorem that formally characterizes how action blending affects system safety; its proof is provided in Appendix D.

Theorem 1. *Under Assumption 1 and Eq.* (6)*, the system in Eq.* (1) *controlled by the blended action in Eq.* (3) *is safe according to Definition 1 if, for some scalar* $\phi > 0$*, the following condition holds:*

$$\mathbf{f}^\top(\mathbf{s}, \mathbf{a}_{\mathsf{learning}}) \cdot \mathbf{M}(\mathbf{s}) \cdot \mathbf{f}(\mathbf{s}, \mathbf{a}_{\mathsf{learning}}) < \frac{1 - (1+\phi) \cdot \alpha \cdot \beta^2(\mathbf{s})}{(1+\phi^{-1}) \cdot (1 - \beta(\mathbf{s}))^2}, \text{ for any } \mathbf{s} \in \mathbb{S}. \tag{8}$$

Theorem 1 characterizes admissible Learning-Student behavior through the right-hand side of Eq. (8), which explicitly depends on the blending weight. We next examine this dependence and derive a critical lower bound on the blending weight for safety assurance. The following corollary summarizes the resulting properties, with its proof provided in Appendix E.

Corollary 1. *For* $0 < \alpha < 1$*,* $0 < \beta < 1$*, and* $\phi > 0$*, define:*

$$g(\beta) \triangleq \frac{1 - (1+\phi) \cdot \alpha \cdot \beta^2}{(1+\phi^{-1}) \cdot (1-\beta)^2}, \text{ and } \beta_- \triangleq \frac{1 + \phi^{-1} - \sqrt{(1+\phi^{-1}) \cdot (1-\alpha)}}{(1+\phi) \cdot \alpha + (1+\phi^{-1})}. \tag{9}$$

Given $(1+\phi) \cdot \alpha < 1$*, we have the following properties: i)* $g(\beta)$ *is monotonically increasing in* β*; ii)* $g(\beta) > 1$ *holds if and only if* $\beta > \beta_-$*; and iii) the infimum of* β_- *is* $\inf_{\phi > 0} \beta_- = \frac{\alpha}{2-\alpha}$*.*

3.2 Safety- and Exploration-Aware Blending

Theorem 1 characterizes how the blending weight governs admissible Learning-Student behavior, while Corollary 1 establishes a critical lower bound on this weight. Combining these results with the exploration measure in Eq. (4), we propose the adaptive blending weight as follows:

$$\textbf{Adaptive Blending Weight: } \beta(\mathbf{s}) = \min\left\{\max\left\{\mathbf{s}^\top \cdot \mathbf{P} \cdot \mathbf{s},\ \frac{\alpha}{2-\alpha}\right\},\ 1\right\} \tag{10}$$

The blending weight in Eq. (10) is guided by three principles derived from the preceding analysis: learning-fault tolerance, safety–exploration balance, and state-dependent adaptation, detailed below.

■ **Learning-Fault Tolerance via a Blending-Weight Lower Bound:** Property ii) of Corollary 1 establishes the critical threshold β_- for blending weight. To understand its significance, we let $\mathbf{s}_{\text{learning}}(k+1) = \mathbf{f}(\mathbf{s}(k), \mathbf{a}_{\text{learning}}(k))$. If $\beta(\mathbf{s}) \leq \beta_-$, then $g(\beta) \leq 1$. Consequently, the safety condition in Eq. (8) would require $\mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{learning}}(k+1) < 1$. By Eq. (6), this implies $\mathbf{s}_{\text{learning}}(k+1) \in \mathbb{S}$. Hence, safety would require the Learning-Student itself to keep the next state within the safety set, leaving no tolerance for learning-induced safety violations. This demonstrates the role of a critical blending threshold in providing learning-fault tolerance.

■ **Safety–Exploration Balance via the Minimum Lower Bound:** Property i) of Corollary 1 shows that increasing β increases $g(\beta)$, thereby allowing greater Learning-Student deviations under Eq. (8). However, Eq. (3) shows that a larger β also reduces the Learning-Student's contribution to the executed action, limiting exploration. This reveals an inherent tradeoff between learning-fault tolerance and exploration. Property iii) further establishes the infimum of the critical threshold (i.e., $\inf_{\phi>0} \beta_- = \frac{\alpha}{2-\alpha}$), which determines the lower-bound term in Eq. (10). Using this bound avoids an unnecessarily restrictive minimum Safety-Teacher contribution, preserving greater exploration scope.

■ **Safety and Exploration Awareness via State-Dependent Adaptiveness:** Eq. (10) adapts $\beta(\mathbf{s})$ according to the state-dependent measure: $\mathbf{s}^\top \cdot \mathbf{P} \cdot \mathbf{s}$. As the system state approaches the boundary of exploration region, this term increases, thereby increasing the contribution of the Safety-Teacher. When $\beta(\mathbf{s}) = 1$, the Learning-Student has zero contribution to the executed action in Eq. (3). Referring to Fig. 2, Eq. (10), and Lemma 6, this occurs for either unsafe states outside $\mathbb{S}$ or safe but unexplored states in $\mathbb{S} \setminus \mathbb{E}$, enabling safety recovery in the former but restricting learning exploration in the latter. To reduce this conservatism, Lemma 6 renders $\mathbb{E}$ the largest-volume exploration region, thereby minimizing $\mathbb{S} \setminus \mathbb{E}$.

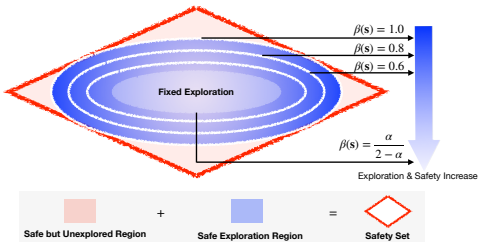


Figure 2: Adaptive blending weight.

4 Learning-Student Module

This section presents a novel design for the Learning-Student, which builds on actor-critic RL to perform runtime learning in the open world and generate the learning action $\mathbf{a}_{\text{learning}}$ in Eq. (3). Efficient runtime learning is highly desirable in open-world environments, with the expectation of reducing learning-induced faults and fast acquiring high-performance action policies for safety- and time-critical missions. However, general-purpose RL frameworks predominantly rely on random batch sampling (Rana et al., 2021; Li et al.; Cheng et al., 2019b; Johannink et al., 2019b; Cheng et al., 2019a; Sasso et al., 2023; Mao et al., 2025; Cao et al., 2024), where replayed samples often misalign with the temporal and state distributions of real-time system states, thereby limiting learning efficiency. In contrast, cognitive studies reveal that humans innately possess efficient learning, enabled by cognition-driven memory retrieval that selectively prioritizes learning experiences according to:

- **Frequency**: Human mind naturally prioritizes familiar and frequently encountered information, facilitating memory retrieval and reducing cognitive effort, thereby enhancing cognitive efficiency (Popov & Reder, 2020; Berglund-Barraza et al., 2019).
- **Recency**: Human memory exhibits a bias toward recently accessed information, enabling faster retrieval and more effective decision-making in complex and non-stationary environments (Sederberg et al., 2008; Holm & Wells, 2023).

- **Similarity**: Human brain uses surface-level and structural cues to retrieve memories, with structural similarity dominating relational knowledge retrieval and supporting efficient generalization (Anderson et al., 1997; 2004; Lebiere et al., 2013; Laughery Jr et al., 2012).

4.1 Cognitive Batch Sampling Mechanism

These above insights from cognitive science motivate us to develop a cognition-inspired experience selection mechanism, termed *cognitive batch sampling*, to enhance runtime learning efficiency. The mechanism consists of the two steps, outlined below.

Step 1: Experience Ranking: We rank experience instances stored in replay buffer according to:

$$\text{rank}(\mathbf{I}_{[i]}, \mathbf{S}_t) = \mathcal{S}(\mathbf{I}_{[i]}, \mathbf{S}_t) + \mathcal{F}(\mathbf{I}_{[i]}) + \mathcal{R}(\mathbf{I}_{[i]}), \quad i \in \{1, 2, \ldots, \mathsf{size}(\mathsf{D})\}, \tag{11}$$

where the $\mathsf{size}(\mathsf{D})$ denotes the size of replay buffer D, the $\mathbf{I}_{[i]}$ is the i-th stored instance in the buffer, and the $\mathcal{S}(\cdot,\cdot), \mathcal{F}(\cdot), \mathcal{R}(\cdot) \in \mathbb{R}$ are measure functions of similarity, frequency, and recency. The $\text{rank}(\cdot,\cdot) \in \mathbb{R}$ ranks experience instances relative to real-time state transition $\mathbf{S}_t$ at current time t:

$$\mathbf{S}_t \triangleq [\, \mathbf{o}(t) \;\mid\; \mathbf{a}(t) \;\mid\; \mathbf{o}(t{+}1) \;\mid\; \mathcal{R}(\mathbf{s}(t), \mathbf{a}(t), \mathbf{s}(t{+}1)) \,], \tag{12}$$

where $\mathbf{o}(\cdot)$ denotes the observation and $\mathcal{R}(\cdot,\cdot,\cdot)$ maps each state–action–next-state to a scalar reward.

Step 2: Experience Retrieval: Given batch size $L \in \mathbb{N}$, we generate sample batch as:

$$\text{Sample Batch: } \mathsf{B} \triangleq \big[\, \mathbf{I}_{[\mathsf{b}_1]};\;\; \mathbf{I}_{[\mathsf{b}_2]};\;\; \ldots;\;\; [\mathbf{I}_{[\mathsf{b}_{L-1}]}];\;\; [\mathbf{I}_{[\mathsf{b}_L]}] \,\big], \tag{13}$$

which are the top-L experience instances drawn from the replay buffer D according to the ranking function in Eq. (11). Specifically, the retrieval process can be mathematically characterized as:

$$\underbrace{\text{rank}(\mathbf{I}_{[\mathsf{b}_1]}, \mathbf{S}_t) \geq \text{rank}(\mathbf{I}_{[\mathsf{b}_2]}, \mathbf{S}_t) \geq \ldots \geq \text{rank}(\mathbf{I}_{[\mathsf{b}_L]}, \mathbf{S}_t)}_{\text{Top-}L\text{ experience instances to be selected to form sample batch}} \geq \text{rank}(\mathbf{I}_{[\mathsf{b}_{L+1}]}, \mathbf{S}_t) \geq \ldots \geq \text{rank}(\mathbf{I}_{[\mathsf{b}_{\mathsf{size}(\mathsf{D})}]}, \mathbf{S}_t).$$

4.2 Measure Functions: Frequency, Recency, and Similarity

The final step in realizing the cognitive batch sampling is the measure functions for frequency, recency, and similarity. In cognitive science, the temporal dynamics model of human declarative memory can simultaneously capture the effects of frequency and recency on memory retrieval (Anderson et al., 2004; Stanley & Byrne, 2016). Building on this model, we propose a variant tailored to runtime learning for autonomous systems with continuous experience spaces, which is formally described in Algorithm 1 to concurrently measure the frequency and recency of the experiences in replay buffer.

Remark 2. *The model in Eq.* (17) *can concurrently capture frequency and recency measures in experience retrieval. Specifically, for the experience instances within cluster indexed by j, a larger N_j indicates a greater number of instances belonging to the cluster j, reflecting a higher frequency of occurrence. Such high-frequency instances consequently receive a higher ranking value in Eq.* (11). *A smaller time elapsed $\Delta_{[j,k]}$ indicates more recently occurring experience instances. Accounting for the memory decay factor $\lambda > 0$, a smaller $\Delta_{[j,k]}$ also results in a higher ranking value in Eq.* (11).

The similarity measure $\mathcal{S}(\cdot,\cdot)$ in Eq. (11) can be readily defined using well-established candidates, such as Gaussian similarity, cosine similarity, and Euclidean-distance similarity.

4.3 Theoretical Analysis

The well-known alternative to random batch sampling is the prioritized experience replay (PER) (Schaul et al., 2016; Fedus et al., 2020), which improves replay efficiency by assigning higher sampling probabilities to transitions with large temporal-difference errors. In contrast, our cognitive sampling aims at addressing **Challenge 2** by retrieving experiences based on their similarity to the current experience, temporal recency, and occurrence frequency. Our theoretical analysis will characterize how these three factors jointly affect contextual mismatch, temporal mismatch, and statistical uncertainty. To facilitate the analysis, we define $\mathbf{g}^\star_{[t]} \triangleq \mathbf{g}(\mathbf{S}_t, t)$ as the ideal gradient associated with real-time experience $\mathbf{S}_t$ under current operating regime. For each experience instance $\mathbf{I}_{[i]}$, we let Δ_i denote its elapsed time, corresponding to $\Delta_{[j,k]}$ in Eq. (17) when $\mathbf{I}_{[i]}$ is the k-th

Algorithm 1 Concurrent Frequency and Recency Measures

Input: Replay buffer D, cluster number $C \in \mathbb{N}$, cluster criterion $\mathsf{c} > 0$.

1: Infuse each experience instance according to:

$$\mathcal{I}(\mathbf{I}_{[i]}) = \left(||\mathbf{o}_{[i]}||_2 + \left\|\mathbf{a}_{[i]}\right\|_2\right) \cdot \exp(\mathcal{R}_{[i]}), \quad i \in \{1, 2, \ldots, \mathsf{size}(\mathsf{D})\} \tag{14}$$

where $\mathbf{o}_{[i]}$, $\mathbf{a}_{[i]}$, and $\mathcal{R}_{[i]}$ denote the observation tensor, action tensor, and reward included in instance $\mathbf{I}_{[i]}$;

2: Identify the cluster index associated with the infused experience instance:

$$\mathcal{C}(\mathbf{I}_{[i]}) = \begin{cases} \lfloor \mathcal{I}(\mathbf{I}_{[i]})/\mathsf{c} \rfloor, & \text{if } \mathcal{I}(\mathbf{I}_{[i]}) < C \cdot \mathsf{c} \\ C, & \text{otherwise} \end{cases}, \tag{15}$$

where $\lfloor \cdot \rfloor$ denotes the floor function;

3: Compute each cluster size:

$$\mathsf{N}_j = \mathsf{sum}(\mathcal{C}(\mathbf{I}_{[i]}) == j), \quad j \in \{0, 1, \ldots, C\} \tag{16}$$

where '==' denotes the equality indicator, which equals 1 if the two operands are equal and 0 otherwise;

4: Compute concurrent frequency and recency measures:

$$\mathcal{F}(\mathbf{I}_{[i]}) + \mathcal{R}(\mathbf{I}_{[i]}) = \log\left(\sum_{k=1}^{\mathsf{N}_j} \Delta_{[j,k]}^{-\lambda}\right), \quad \text{with } \mathcal{C}(\mathbf{I}_{[i]}) = j \in \{0, 1, \ldots, C\} \tag{17}$$

where $\lambda > 0$ is the memory decay factor, and $\Delta_{[j,k]}$ denotes the elapsed time since the k-th activation of the cluster indexed by j, which can be readily read from the storage order of the corresponding experience instances in cluster j within the replay buffer D.

instance associated with cluster j. For each selected experience $\mathbf{I}_{[i]} \in \mathsf{B}$, we model its stochastic gradient and define the corresponding batch-average gradient as:

$$\widehat{\mathbf{g}}_{[i]} = \mathbf{g}(\mathbf{I}_{[i]}, t - \Delta_i) + \boldsymbol{\xi}_{[i]}, \quad \widehat{\mathbf{g}}_{\mathsf{B}} \triangleq \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \frac{\widehat{\mathbf{g}}_{[i]}}{L}. \tag{18}$$

where $\boldsymbol{\xi}_{[i]}$ denotes stochastic gradient noise. By Eqs. (13) and (15) to (17), we further introduce:

$$\rho_{[j]} \triangleq \frac{\max_{1 \le k \le \mathsf{N}_j}\{\Delta_{[j,k]}\}}{\min_{1 \le k \le \mathsf{N}_j}\{\Delta_{[j,k]}\}}, \quad \Delta_{[j]}^{\mathrm{eq}} \triangleq \left(\frac{\mathsf{N}_j}{\sum_{k=1}^{\mathsf{N}_j} \Delta_{[j,k]}^{-\lambda}}\right)^{1/\lambda}, \quad \mathsf{n}_j \triangleq \left|\left\{\mathbf{I}_{[i]} \in \mathsf{B} \;\middle|\; \mathcal{C}(\mathbf{I}_{[i]}) = j\right\}\right|. \tag{19}$$

The following theorem characterizes how the similarity, recency, and frequency jointly affect the gradient error, with detailed proof provided in Appendix F.2.

Theorem 2. *Under Assumptions 2 to 4, the gradient-error bound associated with the actual top-L selected batch, conditioned on the runtime history $\mathcal{H}_t$, satisfies:*

$$\mathbb{E}\left[\left\|\widehat{\mathbf{g}}_{\mathsf{B}} - \mathbf{g}_{[t]}^{\star}\right\|_2 \middle| \mathcal{H}_t\right] \le \underbrace{\frac{L_s}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \frac{1}{\mathcal{S}(\mathbf{I}_{[i]}, \mathbf{S}_t)}}_{\textit{contextual mismatch}} + \underbrace{\frac{L_r}{L} \sum_{\{j : \mathsf{n}_j > 0\}} \mathsf{n}_j \cdot \rho_{[j]} \cdot \Delta_{[j]}^{\mathrm{eq}}}_{\textit{temporal mismatch}} + \frac{\sigma}{\sqrt{L}}. \tag{20}$$

Remark 3. *Equation* (20) *in Theorem 2 reveals that higher similarity reduces contextual mismatch, while greater recency, reflected by a smaller $\Delta_{[j]}^{\mathrm{eq}}$, reduces temporal mismatch. A smaller temporal spread $\rho_{[j]}$ further tightens the temporal-mismatch bound. As further characterized by Corollary 2 in Appendix F.3, occurrence frequency, reflected by n_j, and recency jointly determine the temporal benefit of additional experiences. Specifically, increasing n_j reduces the temporal mismatch only when the additional experiences retain non-negligible recency weights; otherwise, their contribution rapidly diminishes. Collectively, these effects favor similar, recent, and frequently occurring experiences, mitigating experience misalignment and facilitating faster and more stable runtime learning.*

5 Safety-Teacher Module

The Safety-Teacher is a non-learning design dedicated exclusively to safety-critical functions. As shown in Fig. 1, to enable broader safe learning for vision-based missions, its Motion Planner plans

verifiably collision-free motions, while the Controller safely tracks these motions. Given that motion planning (e.g., Interactive-FAR (He et al., 2024)) is well established for the open world, our design focus shifts to the Controller. Autonomous systems usually have environment-dependent dynamics, motivating model adaptation. However, open-world environments cannot be fully characterized by available model knowledge, resulting in unknown model mismatch that requires online estimation and compensation. Accordingly, we propose an adaptive-model-adaptive-control scheme:

$$\mathbf{a}_{\text{safety}}(k) = \mathbf{a}_{\text{model}}(k) + \mathbf{a}_{\text{comp}}(k), \tag{21}$$

where $\mathbf{a}_{\text{model}}(k)$ denotes the model-based action, computed as

$$\mathbf{a}_{\text{model}}(k) = [\mathbf{R}(\mathbf{s}(k)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k))] \cdot \mathbf{s}(k), \tag{22}$$

where $\mathbf{R}(\mathbf{s}(k))$ and $\mathbf{Q}(\mathbf{s}(k))$ can be obtained by solving the following matrix inequalities via CVX (Grant et al., 2009) or LMI (Gahinet et al., 1994) toolbox:

$$\mathbf{I}_p - \mathbf{C} \cdot \mathbf{Q}(\mathbf{s}(k)) \cdot \mathbf{C}^\top \succ 0, \tag{23a}$$

$$\mathbf{Q}(\mathbf{s}(k)) - n \cdot \mathsf{diag}^2(\mathbf{s}(k)) \succ 0, \tag{23b}$$

$$\begin{bmatrix} \widehat{\alpha} \cdot \mathbf{Q}(\mathbf{s}(k)) & * \\ \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{Q}(\mathbf{s}(k)) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{R}(\mathbf{s}(k)) & \mathbf{Q}(\mathbf{s}(k)) \end{bmatrix} \succ 0, \tag{23c}$$

with $0 < \widehat{\alpha} < 1$ and $\mathbf{C}$ given in Eq. (2). Meanwhile, $\mathbf{a}_{\text{comp}}(k)$ in Eq. (21) denotes the compensation action for the unknown model mismatch and is computed as:

$$\mathbf{a}_{\text{comp}}(k) = -\mathbf{B}^\dagger(\mathbf{s}(k)) \cdot \mathbf{h}(\mathbf{s}(k-1)), \tag{24}$$

where $\mathbf{h}(\mathbf{s}(k-1))$ is the previous model mismatch, exactly reconstructed via Eq. (79) to estimate the current mismatch $\mathbf{h}(\mathbf{s}(k))$. For this design, the following theorem establishes the resulting Safety-Teacher property; see Appendix G for its detailed proof.

Theorem 3. *Consider the design of Safety-Teacher, consisting of Eqs.* (21) *to* (24)*. Under Assumption 1, the Safety-Teacher satisfies the design requirement in Eq.* (6)*, if*

$$\widehat{\alpha} + 2\mathbf{s}^\top(k) \cdot \overline{\mathbf{A}}^\top(\mathbf{s}(k)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)) \cdot \Delta\mathbf{h}(k) + \Delta\mathbf{h}^\top(k) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)) \cdot \Delta\mathbf{h}(k) < \alpha, \ \forall \mathbf{s}(k) \in \mathbb{S} \tag{25}$$

and $\mathbf{h}(\mathbf{s}(k-1)) \in \mathsf{range}(\mathbf{B}(\mathbf{s}(k)))$*, where we define:*

$$\overline{\mathbf{A}}(\mathbf{s}(k)) \triangleq \mathbf{A}(\mathbf{s}(k)) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{R}(\mathbf{s}(k)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)), \quad \Delta\mathbf{h}(k) \triangleq \mathbf{h}(\mathbf{s}(k)) - \mathbf{h}(\mathbf{s}(k-1)). \tag{26}$$

Remark 4. *Safety assurance of **OpenRun** relies on the Safety-Teacher satisfying Eq.* (6)*, which Theorem 3 establishes under the condition in Eq.* (25)*. This condition generally holds as the system operates at high sampling frequencies, where the inter-sample model-mismatch variation (i.e.,* $\Delta\mathbf{h}(k)$*) is sufficiently small. However, learning-induced faults can cause abrupt system changes that violate this condition, in which case action blending alone cannot guarantee theoretical safety. As indicated in Remark 3 and experimentally demonstrated in Section 6.1, cognitive batch sampling mitigates this limitation by improving learning efficiency and thereby reducing the occurrence of such faults.*

6 Experiment

This section evaluates **OpenRun** on a quadruped robot and an off-road autonomous vehicle in real and simulated wild forests. **Note: Supplementary Material also include field-experiment videos**.

6.1 Open-World Experiment: Real Wild Forests

This section presents experiment on runtime learning of quadruped robot for navigation in real wild forests. As shown in Fig. 6, these open-world environments exhibit substantial complexity and diversity, featuring unstructured and uneven terrain, randomly distributed trees and dense vegetation, dead zones, swales, leaf-covered holes, and rocky paths with rounded, densely distributed cobblestones. Appendix H in appendix presents the detailed experiment design.

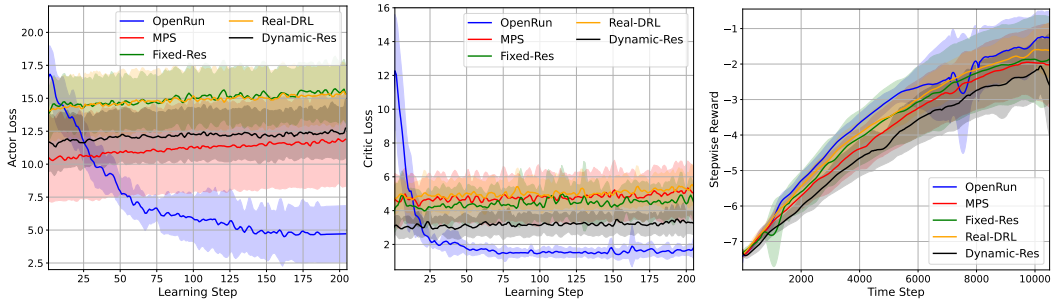
6.1.1 Comparative Studies

Two major classes of safe RL frameworks have emerged that can potentially enable runtime learning in the open world: 1) Runtime-Assurance RL, including model predictive shielding (MPS) (Banerjee et al., 2024) and Real-DRL (Mao et al., 2025); and 2) Residual RL, including fixed residual learning (Fixed-Res) (Cao et al., 2024; Johannink et al., 2019a) and dynamic residual learning (Dynamic-Res) (Cao et al., 2026). We comprehensively compare these approaches with our **OpenRun**.

We perform 15 independent trips of runtime learning for navigation. Their trajectories of actor loss, critic loss, and stepwise reward are shown in Fig. 3. The quantitative safety performance is summarized in Table 1. In addition, the robot behaviors (3× speed) during uniformly sampled trips (1, 8, and 15) are available via [runtime-learning-comparison link]. These results demonstrate that, under the worst-case safety conditions in Eq. (90), **OpenRun** enables safer, more stable, and faster runtime learning with higher task performance than state-of-the-art safe RL frameworks in challenging wild-forest environments.

Method ID	Balance Loss	Collision	Dead-Zone Encounter	Task Completion
OpenRun	1/15	1/15	0/15	13/15
Real-DRL	3/15	2/15	2/15	8/15
MPS	5/15	4/15	0/15	6/15
Fixed-Res	1/15	4/15	1/15	9/15
Dynamic-Res	7/15	1/15	0/15	7/15

Table 1: Statistical safety comparison of runtime learning across different methods. **Note:** "Task Completion = No Collision + No Loss of Balance + No Dead-Zone Encounter + Goal Reached." Any collision, balance loss, or dead-zone encounter immediately terminates the runtime learning trial to protect robot from damage.



(a) Actor Loss (95% CI). (b) Critic Loss (95% CI). (c) Stepwise Reward (95% CI).

Figure 3: Comparison of runtime learning performance between **OpenRun** and baseline methods.

6.1.2 Ablation Studies

The **OpenRun** incorporates two core innovations: i) adaptive action blending (denoted as 'Blending') and ii) cognitive batch sampling (denoted as 'Cognitive'). Without either innovation, **OpenRun** can reduce to residual RL with random batch sampling (denoted as 'Residual-Random'). Accordingly, to evaluate the individual contributions of these innovations, we compare three variants: 1) 'Blending-Cognition', 2) 'Blending-Random', and 3) 'Residual-Random'.

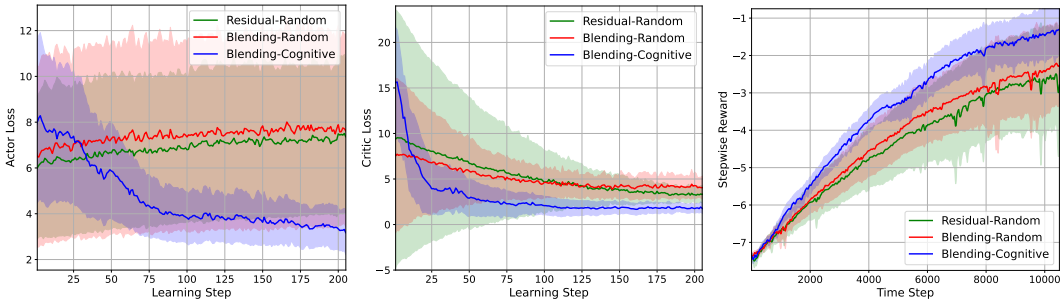
The trajectories of actor loss, critic loss, and stepwise reward over 15 independent trips of runtime learning are shown in Fig. 4. Their quantitative safety performance is summarized in Table 2. In addition, the robot behaviors (3× speed) during uniformly sampled trips (1, 8, and 15) are available via [runtime-learning-ablation link]. These results reveal the complementary roles of the two innovations. Specifically, adaptive blending substantially improves safety assurance over Residual-Random, although Blending-Random still exhibits learning-induced faults. This observation is consistent with Remark 4: the practical conditions underlying the theoretical safety assurance may occasionally be violated during runtime learning. Cognitive batch sampling further reduces learning-induced faults by mitigating experience misalignment, as demonstrated by the substantially fewer failures of Blending-Cognition, although occasional failures remain.

Method ID	Balance Loss	Collision	Dead-Zone Encounter	Task Completion
Blending-Cognition	1/15	1/15	0/15	13/15
Blending-Random	5/15	1/15	3/15	6/15
Residual-Random	11/15	0/15	1/15	3/15

Table 2: Statistical safety performance.

6.1.3 Cross-Environment Generalization

We further demonstrate **OpenRun**'s capability for cross-environment generalization. As shown in Fig. 6, the robot performs runtime learning across three distinct environments. Specifically, the robot first completes 15 trips of runtime learning in Environment 1. Without resetting the learned policy or accumulated experience, the same **OpenRun** instance is then directly deployed in the



(a) Actor Loss (95% CI). (b) Critic Loss (95% CI). (c) Stepwise Reward (95% CI).

Figure 4: Ablation with respect to the core innovations: adaptive blending and cognitive sampling.

previously unseen Environment 2 and subsequently Environment 3, where runtime learning continues from the transferred learning state. The robot's learning behaviors (2× speed) are available via [runtime-learning-environment link], demonstrating that **OpenRun** maintains safe runtime learning under sequential cross-environment transfer without reinitialization.

6.2 Open-World Experiment: Simulated Wild Forests

This section presents an experiment in a simulated open world; see Appendix I for its design.

To demonstrate the generalizability of **OpenRun** across different platforms and functions, we further deploy **OpenRun** on an off-road autonomous vehicle to perform runtime learning for navigation planning. As shown in Fig. 7, the operating environment features randomized obstacles (rocks and trees) over terrains ranging from flat to uneven surfaces. This experiment extends the ablation study in Section 6.1.2 by further examining the three factors in the proposed cognitive batch sampling: frequency, recency, and similarity. Since the measure in Eq. (17) jointly captures frequency and recency, we consider three variants for comparison: 1) frequency and recency only (denoted as 'F-R'), 2) similarity only (denoted as 'S'), and 3) all three factors (denoted as 'F-R-S'). When these three cognitive factors are removed, we reduce sampling strategy to prioritized experience replay (denoted as 'PER'), rather than random sampling. Figure 5 presents the episode-average rewards, while the corresponding actor and critic losses are shown in Fig. 8. The results show that jointly incorporating all three cognitive factors enables safer, faster, and more stable runtime learning with higher task performance, further supporting the theoretical result in Theorem 2. In contrast, PER provides limited benefit in this safety-critical runtime-learning setting.

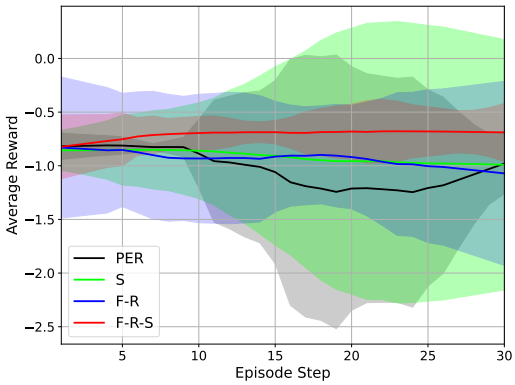


Figure 5: Stepwise rewards (95% CI) over five random seeds, each with 30 episodes.

The non-learning design of the Safety-Teacher in Section 5 is also a contribution of this paper. Due to space limitations, its evaluation on a locomotion-control task is presented in Appendix I.3.

7 Conclusion and Discussion

This paper introduces the **OpenRun** framework, enabling runtime learning for safety-critical autonomous systems in open-world environments. Its core innovations lie in safety- and exploration-aware action blending and cognitive batch sampling, which jointly enhance safety assurance and learning efficiency. Comprehensive experiments on a real quadruped robot and an off-road autonomous vehicle demonstrate the distinctive capabilities of **OpenRun** across different platforms and tasks, including runtime safety assurance, experience efficiency, and task performance. Overall, **OpenRun** provides a unified framework that reconciles safety assurance with continuous learning and adaptation in open-world environments.

AI USE STATEMENT

In this work, we used generative AI tools to assist with refining hypothesis. We did not use generative AI tools to generate synthetic datasets, design experiments, implement the proposed methods, analyze experimental data, or interpret experimental results, and the rest of tasks requiring disclosure are not applicable to this work. Additionally, we used generative AI tools to improve the readability and presentation of the manuscript, including editing and polishing text, editing software code, and refining the organization and structure of technical descriptions. We have reviewed all AI-assisted work. We take full responsibility for the final content of this work, including all text, claims, mathematical results, and other artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies runtime learning and safety assurance for autonomous systems operating in open-world environments. The research does not involve human subjects, personally identifiable information, or sensitive personal data. The experiments are conducted using robotic systems in simulation and controlled physical environments and therefore do not require human-subject approval. We are not aware of any additional ethical concerns, conflicts of interest, or issues related to discrimination, privacy, or data ownership arising from this work. We have aimed to report the methodology, theoretical assumptions, experimental settings, and results transparently and accurately to support reproducibility and responsible use of the research.

REPRODUCIBILITY STATEMENT

We have made efforts to facilitate the reproducibility of both the theoretical and experimental results presented in this work. The main paper provides detailed descriptions of the proposed **OpenRun** framework and its key components. The experimental section describes the simulation and real-world robotic platforms, evaluation settings, baselines, and performance metrics, while additional implementation details, hyperparameters, and experimental configurations are provided in the paper. Source code and configuration files necessary to reproduce the experiments are included in the anonymous supplementary materials.

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A Default Notation

The following standardized notation is used throughout the paper.

a	A scalar
$\mathbf{a}$	A vector
$\mathbf{A}$	A matrix
$\mathbb{A}$	A set
$\mathbf{1}_m$	m-dimensional vectors with entries of all ones
$\mathbf{0}_m$	m-dimensional vectors with entries of all zeros
$\mathbf{I}_m$	Identity matrix with m rows and m columns
$\top$	Matrix transposition
$\cdot$	Multiplication operator
$*$	Transpose of the corresponding off-diagonal block
$\mathsf{diag}(\mathbf{a})$	A square, diagonal matrix with diagonal entries given by elements of $\mathbf{a}$
$\log()$	Logarithm function
$\det()$	Determinant of a matrix
$\mathbf{A}^{-1}$	Inverse of matrix $\mathbf{A}$
$\mathbb{R}^n$	Set of n-dimensional real vectors
$\mathbb{N}$	Set of natural numbers
$\mathbf{c} > \mathbf{0}_n$	All entries of $\mathbf{c} \in \mathbb{R}^n$ are strictly positive
$\mathbf{P} \succ 0$	Matrix $\mathbf{P}$ is positive definite
$\mathbf{B}^\dagger$	Moore–Penrose pseudoinverse of matrix $\mathbf{B}$
$\mathsf{range}(\mathbf{B})$	Column space of matrix $\mathbf{B}$.

B AUXILIARY EXAMPLES

Through appropriate state representations, the conditions defining the safety set in Eq. (2) can also equivalently characterize a broad class of nonlinear safety conditions. We now provide several concrete examples:

- **Slip Regulation**. For autonomous vehicles, to prevent slipping and skidding, the slip ratio during both traction and braking must remain below a prescribed threshold (Rajamani, 2011), which can be mathematically expressed as

$$\left| \frac{R\omega - v_x}{\max\left(|R\omega|, |v_x|\right)} \right| \leq \lambda_{\max}, \tag{27}$$

  where R denotes the effective wheel radius, ω is the wheel angular velocity, and v_x is the longitudinal vehicle velocity. Under the traction condition, $R\omega \geq v_x$, and the Eq. (27) is thus equivalent to the linear $(1 - \lambda_{\max})R\omega - v_x \leq 0$. Under the braking condition, $v_x \geq R\omega$, yielding the linear $(1 - \lambda_{\max})v_x - R\omega \leq 0$. Therefore, although the original slip-ratio constraint in Eq. (27) is nonlinear, it can be equivalently represented by linear safety conditions in the form of $\mathbf{C} \cdot \mathbf{s} \leq \mathbf{1}_p$ used to define the safety set in Eq. (2).

- **Collision Avoidance**. For mobile robotic systems, collision avoidance requires the distance between the system and surrounding obstacles to remain above a prescribed safety threshold, which can be expressed as

$$||\mathbf{p} - \mathbf{p}_o||_2 \geq d_{\min}, \tag{28}$$

  where $\mathbf{p}$ and $\mathbf{p}_o$ denote the positions of the autonomous system and obstacle, respectively, and $d_{\min}$ denotes the minimum safe distance. By defining the safety-related state $s_d \triangleq d_{\min}/||\mathbf{p} - \mathbf{p}_o||_2$, Eq. (28) is equivalently transformed into the linear condition $s_d \leq 1$, which can be directly represented in the form of $\mathbf{C} \cdot \mathbf{s} \leq \mathbf{1}_p$ used to define the safety set in Eq. (2).

- **Friction-Limit Regulation**. For autonomous vehicles, the resultant tire force must remain within the tire–road friction limit to prevent loss of traction and lateral stability, which can be expressed as

$$\sqrt{F_x^2 + F_y^2} \leq \mu F_z, \tag{29}$$

  where F_x, F_y, and F_z denote the longitudinal, lateral, and normal tire forces, respectively, and μ denotes the tire–road friction coefficient. By defining the safety-related state as $s_\mu \triangleq \frac{\sqrt{F_x^2+F_y^2}}{\mu F_z}$, the nonlinear safety condition in Eq. (29) can be equivalently transformed into the linear condition $s_\mu \leq 1$, which can be directly represented in the form of $\mathbf{C} \cdot \mathbf{s} \leq \mathbf{1}_p$ used to define the safety set in Eq. (2).

C AUXILIARY LEMMAS

The auxiliary lemmas that will be used in the proofs of the main results of this article.

Lemma 1 (Schur Complement (Zhang, 2006)). *For any symmetric matrix* $\mathbf{M} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{bmatrix}$, $\mathbf{M} \succ 0$ *if and only if* $\mathbf{C} \succ 0$ *and* $\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\top \succ 0$.

Lemma 2. *(Seto & Sha, 1999, Lemma 4.1). Consider the safety set* $\mathbb{S}$ *defined in Eq.* (2) *and the exploration set* $\mathbb{E}$ *defined in Eq.* (4)*. Then,* $\mathbb{E} \subseteq \mathbb{S}$ *if* $\mathbf{I}_p - \mathbf{C} \cdot \mathbf{P}^{-1} \cdot \mathbf{C}^\top \succ 0$.

Lemma 3. *(Boyd & Vandenberghe, 2004, Sec. 8.4). For set* $\mathbb{E} = \left\{\mathbf{s} \in \mathbb{R}^n \,\middle|\, \mathbf{s}^\top \cdot \mathbf{P} \cdot \mathbf{s} \leq 1, \mathbf{P} \succ 0\right\}$*, its volume is given by* $\mathrm{vol}(\mathbb{E}) = \mathrm{vol}(\mathbb{B}_n) \cdot \det(\mathbf{P})^{-1/2}$*, where* $\mathbb{B}_n$ *denotes the* n*-dimensional unit Euclidean ball, and* $\mathrm{vol}(\mathbb{B}_n)$ *depends only on the dimension* n*.*

Lemma 4. *(Horn & Johnson, 2013, Sec. 7.7). Given* $\mathbf{A} \succ 0$ *and* $\mathbf{B} \succ 0$*,* $\mathbf{A} \succ \mathbf{B}$ *if and only if* $\mathbf{B}^{-1} \succ \mathbf{A}^{-1}$.

Lemma 5. *(Penrose, 1955, Theorem 1) For any matrix* $\mathbf{B} \in \mathbb{R}^{n \times m}$*, its Moore–Penrose pseudoinverse* $\mathbf{B}^\dagger$ *exists uniquely and satisfies* $\mathbf{B} \cdot \mathbf{B}^\dagger \cdot \mathbf{B} = \mathbf{B}$*. Consequently, for every* $\mathbf{v} \in \mathsf{range}(\mathbf{B})$*, we have* $\mathbf{B} \cdot \mathbf{B}^\dagger \cdot \mathbf{v} = \mathbf{v}$.

Lemma 6. *The safety set* $\mathbb{S}$ *defined in Eq.* (2) *and the exploration region* $\mathbb{E}$ *defined in Eq.* (4) *have the following properties: i)* $\mathbb{E} \subset \mathbb{S}$*; and ii)* $\mathbb{E}$ *has the largest origin-centered volume.*

Proof. The conditions in Eq. (5) ensure that the computed $\mathbf{P}$ in Eq. (2) satisfies $\mathbf{I}_p - \mathbf{C} \cdot \mathbf{P}^{-1} \cdot \mathbf{C}^\top \succ 0$. By Lemma 2, this implies $\mathbb{E} \subset \mathbb{S}$. Meanwhile, by Lemma 3, the volume of $\mathbb{E}$ is proportional to $\sqrt{\det(\mathbf{P}^{-1})}$. Therefore, maximizing the volume of $\mathbb{E}$ is equivalent to maximizing $\det(\mathbf{P}^{-1})$, as formulated in Eq. (5). Hence, the resulting exploration region $\mathbb{E}$ has the largest volume. □

D PROOF OF THEOREM 1

From the affine dynamics in Eq. (7), we define:

$$\mathbf{s}_{\text{learning}}(k+1) \triangleq \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\text{learning}}(k) + \mathbf{h}(\mathbf{s}(k)), \tag{30}$$

$$\mathbf{s}_{\text{safety}}(k+1) \triangleq \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\text{safety}}(k) + \mathbf{h}(\mathbf{s}(k)), \tag{31}$$

where $\mathbf{s}_{\text{learning}}(k+1)$ and $\mathbf{s}_{\text{safety}}(k+1)$ denote the next states resulting from the sole control of the Learning-Student and Safety-Teacher, respectively.

Substituting the blended action in Eq. (3) into the affine dynamics in Eq. (7) yields:

$$\begin{aligned}\mathbf{s}(k+1) &= \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot (\beta(\mathbf{s}(k)) \cdot \mathbf{a}_{\text{safety}}(k) + (1-\beta(\mathbf{s}(k))) \cdot \mathbf{a}_{\text{learning}}(k)) + \mathbf{h}(\mathbf{s}(k)) \\ &= \beta(\mathbf{s}(k)) \cdot (\mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\text{safety}}(k) + \mathbf{h}(\mathbf{s}(k))) \\ &\quad + (1 - \beta(\mathbf{s}(k))) \cdot (\mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\text{learning}}(k) + \mathbf{h}(\mathbf{s}(k))).\end{aligned} \tag{32}$$

Using the definitions in Eqs. (30) and (31), Eq. (32) simplifies to:

$$\mathbf{s}(k+1) = \beta(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}(k+1) + (1 - \beta(\mathbf{s}(k))) \cdot \mathbf{s}_{\text{learning}}(k+1),$$

based on which, expanding the corresponding quadratic form yields:

$$\begin{aligned}&\mathbf{s}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k+1) \\ &= \beta^2(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}(k+1) \\ &\quad + (1 - \beta(\mathbf{s}(k)))^2 \cdot \mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{learning}}(k+1) \\ &\quad + 2\beta(\mathbf{s}(k)) \cdot (1 - \beta(\mathbf{s}(k))) \cdot \mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}(k+1).\end{aligned} \tag{33}$$

Applying Young's inequality with any scalar $\phi > 0$ to the cross term in Eq. (33) gives:

$$\begin{aligned}&2\beta(\mathbf{s}(k)) \cdot (1 - \beta(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}(k+1) \\ &\leq \phi \cdot \beta^2(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}(k+1) \\ &\qquad + \phi^{-1} \cdot (1 - \beta(\mathbf{s}(k)))^2 \cdot \mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{learning}}(k+1).\end{aligned} \tag{34}$$

Combining Eqs. (33) and (34) yields:

$$\begin{aligned}&\mathbf{s}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k+1) \\ &\leq (1+\phi) \cdot \beta^2(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{safety}}(k+1) \\ &\qquad + (1+\phi^{-1}) \cdot (1 - \beta(\mathbf{s}(k)))^2 \cdot \mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{learning}}(k+1),\end{aligned} \tag{35}$$

Applying the Safety-Teacher design requirement in Eq. (6) to Eq. (35) further gives:

$$\begin{aligned}&\mathbf{s}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k+1) \\ &\leq (1+\phi) \cdot \alpha \cdot \beta^2(\mathbf{s}(k)) + (1+\phi^{-1}) \cdot (1 - \beta(\mathbf{s}(k)))^2 \cdot \mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{learning}}(k+1).\end{aligned}$$

Therefore, a sufficient condition for $\mathbf{s}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k+1) < 1$ is:

$$\begin{aligned}&(1+\phi) \cdot \alpha \cdot \beta^2(\mathbf{s}(k)) \\ &\qquad + (1+\phi^{-1}) \cdot (1 - \beta(\mathbf{s}(k)))^2 \cdot \mathbf{s}_{\text{learning}}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\text{learning}}(k+1) < 1.\end{aligned} \tag{36}$$

Restoring the explicit time index k and using the definition of $\mathbf{s}_{\text{learning}}(k+1)$ in Eq. (30), the condition in Eq. (36) is equivalent to that in Eq. (8). Hence, whenever Eq. (8) holds, we have $\mathbf{s}^\top(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k+1) < 1$, which, in conjunction with Eq. (6), implies $\mathbf{s}(k+1) \in \mathbb{S}$. Since the above argument holds for any $\mathbf{s}(k) \in \mathbb{S}$, it can be applied recursively. Therefore, starting from any $\mathbf{s}(0) \in \mathbb{S}$, we have $\mathbf{s}(k) \in \mathbb{S}$ for all $k \in \mathbb{N}$, which establishes the safety guarantee in Definition 1 and completes the proof.

E PROOF OF COROLLARY 1

E.1 PROOF OF PROPERTY I)

For $g(\beta)$ defined in Eq. (9), its derivative with respect to β is:

$$\frac{\partial g(\beta)}{\partial \beta} = \frac{2(1-(1+\phi)\cdot\alpha\cdot\beta)}{(1+\phi^{-1})\cdot(1-\beta)^3}. \tag{37}$$

Since $\phi > 0$ and $0 < \beta < 1$, the denominator in Eq. (37) is strictly positive. Moreover, because $\beta < 1$, we have $1-(1+\phi)\cdot\alpha\cdot\beta > 1-(1+\phi)\cdot\alpha$. Therefore, under $(1+\phi)\cdot\alpha < 1$, we have $1-(1+\phi)\cdot\alpha\cdot\beta > 0$, and hence $\frac{\partial g(\beta)}{\partial \beta} > 0$. Thus, $g(\beta)$ is strictly monotonically increasing in β, which proves Property i).

E.2 PROOF OF PROPERTY II)

Since $0 < \beta < 1$ and $\phi > 0$, we have $(1+\phi^{-1})\cdot(1-\beta)^2 > 0$. Therefore, $g(\beta) > 1$ is equivalent to:

$$1-(1+\phi)\cdot\alpha\cdot\beta^2 > (1+\phi^{-1})\cdot(1-\beta)^2. \tag{38}$$

Expanding and rearranging Eq. (38) leads to the conclusion that $g(\beta) > 1$ if and only if:

$$((1+\phi)\cdot\alpha + 1 + \phi^{-1})\cdot\beta^2 - 2(1+\phi^{-1})\cdot\beta + \phi^{-1} < 0. \tag{39}$$

We define:

$$a \triangleq (1+\phi)\cdot\alpha + 1 + \phi^{-1}, \qquad b \triangleq 1+\phi^{-1}. \tag{40}$$

Then, Eq. (39) becomes $a\cdot\beta^2 - 2b\cdot\beta + \phi^{-1} < 0$, whose two roots are:

$$\beta_\pm = \frac{b \pm \sqrt{b^2 - a\cdot\phi^{-1}}}{a}. \tag{41}$$

Considering Eq. (40), the discriminant term satisfies:

$$b^2 - a\cdot\phi^{-1} = (1+\phi^{-1})^2 - \phi^{-1}\cdot((1+\phi)\cdot\alpha + 1 + \phi^{-1}) = (1+\phi^{-1})\cdot(1-\alpha),$$

substituting which and Eq. (40) into Eq. (41) yields:

$$\beta_\pm = \frac{1+\phi^{-1} \pm \sqrt{(1+\phi^{-1})\cdot(1-\alpha)}}{(1+\phi)\cdot\alpha + 1 + \phi^{-1}}. \tag{42}$$

We next show that $\beta_+ \geq 1$ under $(1+\phi)\cdot\alpha < 1$. From Eq. (42), the $\beta_+ > 1$ is equivalent to:

$$\sqrt{(1+\phi^{-1})\cdot(1-\alpha)} > (1+\phi)\cdot\alpha. \tag{43}$$

Since both sides in Eq. (43) are nonnegative, squaring both sides gives:

$$(1+\phi^{-1})\cdot(1-\alpha) > (1+\phi)^2\cdot\alpha^2,$$

which is further equivalent to:

$$\phi\cdot(1+\phi)\cdot\alpha^2 + \alpha - 1 < 0. \tag{44}$$

Noting that $\phi\cdot(1+\phi)\cdot\alpha^2 + \alpha - 1 = ((1+\phi)\cdot\alpha - 1)\cdot(\phi\cdot\alpha + 1)$ and $\phi\cdot\alpha + 1 > 0$, condition $(1+\phi)\cdot\alpha < 1$ implies Eq. (44). Therefore, our conclusion is:

$$\beta_+ > 1, \text{ if } (1+\phi)\cdot\alpha < 1. \tag{45}$$

Moreover, $(1+\phi)\cdot\alpha < 1$ implies $\alpha < 1$, so the quadratic in Eq. (39) has two distinct real roots. Since its leading coefficient is positive, Eq. (39) holds if and only if $\beta_- < \beta < \beta_+$. Combining this result with $0 < \beta < 1$ and Eq. (45), we arrive at the conclusion that given $(1+\phi)\cdot\alpha < 1$, if and only if:

$$\beta > \beta_- = \frac{1+\phi^{-1} - \sqrt{(1+\phi^{-1})\cdot(1-\alpha)}}{(1+\phi)\cdot\alpha + (1+\phi^{-1})},$$

we have $g(\beta) > 1$. This proves Property ii).

E.3 PROOF OF PROPERTY III)

The lower root β_- in Eq. (42) can be equivalently written as:

$$\beta_- = \frac{1 - \sqrt{(1-\alpha) \cdot \frac{\phi}{1+\phi}}}{1 + \alpha \cdot \phi}. \tag{46}$$

Defining $x \triangleq \sqrt{\frac{\phi}{1+\phi}}$, we have $\phi = \frac{x^2}{1-x^2}$. Meanwhile, we let $c \triangleq \sqrt{1-\alpha}$. Substituting these expressions into Eq. (46) gives $\beta_- = \frac{1-x^2}{1+c\cdot x}$. Differentiating it with respect to x yields $\frac{d\beta_-}{dx} = -\frac{c+2x+c\cdot x^2}{(1+c\cdot x)^2} < 0$, indicating the β_- is strictly decreasing in x. Since $x = \sqrt{\frac{\phi}{1+\phi}}$ is strictly increasing in $\phi > 0$, the β_- is also strictly decreasing in ϕ. Under $(1+\phi)\cdot\alpha < 1$, the admissible range of ϕ is $0 < \phi < \frac{1-\alpha}{\alpha}$. Therefore, the infimum of β_- over all admissible ϕ is attained in the limit as ϕ approaches its upper bound:

$$\inf_{\substack{\phi>0 \\ (1+\phi)\cdot\alpha<1}} \beta_- = \lim_{\phi \to \left(\frac{1-\alpha}{\alpha}\right)^-} \beta_- = \frac{\alpha}{2-\alpha},$$

which completes the proof of Property iii).

F THEOREM 2 AND COROLLARY

This section first provides the assumptions of Theorem 2, then its proof and corollary.

F.1 ASSUMPTIONS

The assumptions underlying Theorem 2 are outlined below.

Assumption 2 (Contextual Smoothness)**.** *We let $\mathbf{g}(\mathbf{I}, t)$ denote the expected gradient associated with experience $\mathbf{I}$ under the operating regime at time t. For any experiences $\mathbf{I}$ and $\mathbf{J}$, we have:*

$$\|\mathbf{g}(\mathbf{I}, t) - \mathbf{g}(\mathbf{J}, t)\|_2 \leq \frac{L_s}{\mathcal{S}(\mathbf{I}, \mathbf{J})},$$

where $L_s > 0$ and $\mathcal{S}(\cdot, \cdot) > 0$ is the similarity measure used in Eq. (11)*.*

Assumption 3 (Temporal Smoothness)**.** *The expected gradient has bounded temporal variation. Specifically, for any experience $\mathbf{I}$ and any t_1, t_2, we have:*

$$\|\mathbf{g}(\mathbf{I}, t_1) - \mathbf{g}(\mathbf{I}, t_2)\|_2 \leq L_r \cdot |t_1 - t_2|,$$

where $L_r > 0$ characterizes the temporal variation of the operating regime.

Assumption 4 (Stochastic-Gradient Noise)**.** *Conditioned on the runtime history $\mathcal{H}_t$, the retrieved batch B and its corresponding elapsed ages are fixed. For each selected experience $\mathbf{I}_{[i]} \in \mathsf{B}$, the stochastic-gradient noise $\boldsymbol{\xi}_{[i]}$ in Eq.* (18) *satisfies: $\mathbb{E}\left[\boldsymbol{\xi}_{[i]} \mid \mathcal{H}_t\right] = \mathbf{0}$, $\mathbb{E}\left[\left\|\boldsymbol{\xi}_{[i]}\right\|_2^2 \mid \mathcal{H}_t\right] \leq \sigma^2$, and $\mathbb{E}\left[\boldsymbol{\xi}_{[i]}^\top \boldsymbol{\xi}_{[\ell]} \mid \mathcal{H}_t\right] = 0$ for $i \neq \ell$, where $\sigma > 0$ is finite.*

F.2 PROOF OF THEOREM 2

This section provides the detailed proof of Theorem 2, which is separated into two steps.

F.2.1 STEP 1: BOUNDING THE AVERAGE ELAPSED AGE OF THE SELECTED BATCH

We first establish a bound on the elapsed ages of the experiences selected into the sample batch.

For a cluster j, we define:

$$\Delta_{j,\min} \triangleq \min_{1 \leq k \leq \mathsf{N}_j} \{\Delta_{[j,k]}\}, \quad \text{and} \quad \Delta_{j,\max} \triangleq \max_{1 \leq k \leq \mathsf{N}_j} \{\Delta_{[j,k]}\},$$

which, in light of the definition of $\rho_{[j]}$ in Eq. (19), leads to:

$$\rho_{[j]} = \frac{\Delta_{j,\max}}{\Delta_{j,\min}}. \tag{47}$$

Since $\lambda > 0$, for every $k \in \{1, \ldots, \mathsf{N}_j\}$, we have:

$$\Delta_{j,\max}^{-\lambda} \leq \Delta_{[j,k]}^{-\lambda} \leq \Delta_{j,\min}^{-\lambda},$$

summing which over all N_j experiences in cluster j gives:

$$\mathsf{N}_j \cdot \Delta_{j,\max}^{-\lambda} \leq \sum_{k=1}^{\mathsf{N}_j} \Delta_{[j,k]}^{-\lambda} \leq \mathsf{N}_j \cdot \Delta_{j,\min}^{-\lambda}. \tag{48}$$

Because all the terms in Eq. (48) are positive, taking reciprocals and then the positive $1/\lambda$ power yields:

$$\Delta_{j,\min} \leq \left(\frac{\mathsf{N}_j}{\sum_{k=1}^{\mathsf{N}_j} \Delta_{[j,k]}^{-\lambda}} \right)^{1/\lambda} \leq \Delta_{j,\max}. \tag{49}$$

Using the definition of $\Delta_{[j]}^{\mathrm{eq}}$ in Eq. (19), we obtain from Eq. (49) that:

$$\Delta_{j,\min} \le \Delta_{[j]}^{\mathrm{eq}} \le \Delta_{j,\max}. \tag{50}$$

Considering Eqs. (47) and (50), for any selected experience belonging to cluster j, its elapsed age satisfies:

$$\Delta_{[j,k]} \le \Delta_{j,\max} = \rho_{[j]} \cdot \Delta_{j,\min} \le \rho_{[j]} \cdot \Delta_{[j]}^{\mathrm{eq}}. \tag{51}$$

Since Δ_i denotes the elapsed time of $\mathbf{I}_{[i]}$, the selected experiences can be equivalently indexed by their cluster assignments. Specifically, for each cluster j, we define:

$$\mathbb{K}_{[j]}^{\mathcal{B}} \triangleq \left\{ k \in \{1, \dots, \mathsf{N}_j\} \;\middle|\; \mathbf{I}_{[j,k]} \in \mathsf{B} \right\}, \tag{52}$$

where $\mathbf{I}_{[j,k]}$ denotes the k-th instance associated with cluster j. By the definition of n_j in Eq. (19), we have:

$$\left| \mathbb{K}_{[j]}^{\mathsf{B}} \right| = \mathsf{n}_j. \tag{53}$$

From Eqs. (51) and (53), we further have:

$$\begin{aligned} \frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \Delta_i &= \frac{1}{L} \sum_{\{j:\mathsf{n}_j > 0\}} \sum_{k \in \mathbb{K}_{[j]}^{\mathsf{B}}} \Delta_{[j,k]} \le \frac{1}{L} \sum_{\{j:\mathsf{n}_j > 0\}} \sum_{k \in \mathbb{K}_{[j]}^{\mathsf{B}}} \Delta_{j,\max} = \frac{1}{L} \sum_{\{j:\mathsf{n}_j > 0\}} \left| \mathbb{K}_{[j]}^{\mathsf{B}} \right| \Delta_{j,\max} \\ &= \frac{1}{L} \sum_{\{j:\mathsf{n}_j > 0\}} \mathsf{n}_j \cdot \Delta_{j,\max} \le \sum_{\{j:\mathsf{n}_j > 0\}} \frac{\mathsf{n}_j}{L} \cdot \rho_{[j]} \cdot \Delta_{[j]}^{\mathrm{eq}}. \end{aligned} \tag{54}$$

F.2.2 Step 2: Bounding the Batch Gradient Error

We next derive the gradient-error bound. From Eq. (18) and $\mathbf{g}_{[t]}^{\star} = \mathbf{g}(\mathbf{S}_t, t)$, we have:

$$\widehat{\mathbf{g}}_{\mathsf{B}} - \mathbf{g}_{[t]}^{\star} = \frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \left[\mathbf{g}(\mathbf{I}_{[i]}, t - \Delta_i) - \mathbf{g}(\mathbf{S}_t, t) \right] + \frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \boldsymbol{\xi}_{[i]},$$

which is equivalent to:

$$\begin{aligned} &\widehat{\mathbf{g}}_{\mathsf{B}} - \mathbf{g}_{[t]}^{\star} \\ &= \underbrace{\frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \left[\mathbf{g}(\mathbf{I}_{[i]}, t) - \mathbf{g}(\mathbf{S}_t, t) \right]}_{\triangleq \mathbf{e}_{\mathsf{context}}} + \underbrace{\frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \left[\mathbf{g}(\mathbf{I}_{[i]}, t - \Delta_i) - \mathbf{g}(\mathbf{I}_{[i]}, t) \right]}_{\triangleq \mathbf{e}_{\mathsf{temporal}}} + \underbrace{\frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \boldsymbol{\xi}_{[i]}}_{\triangleq \mathbf{e}_{\mathsf{noise}}}. \end{aligned} \tag{55}$$

We next bound the three terms in Eq. (55) separately.

First, by the triangle inequality and Assumption 2, we have:

$$\left\| \mathbf{e}_{\mathsf{context}} \right\|_2 \le \frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \left\| \mathbf{g}(\mathbf{I}_{[i]}, t) - \mathbf{g}(\mathbf{S}_t, t) \right\|_2 \le \frac{L_s}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \frac{1}{\mathcal{S}(\mathbf{I}_{[i]}, \mathbf{S}_t)}. \tag{56}$$

Second, by the triangle inequality and Assumption 3, we have:

$$\begin{aligned} \left\| \mathbf{e}_{\mathsf{temporal}} \right\|_2 \le \frac{1}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \left\| \mathbf{g}(\mathbf{I}_{[i]}, t - \Delta_i) - \mathbf{g}(\mathbf{I}_{[i]}, t) \right\|_2 &\le \frac{L_r}{L} \sum_{\mathbf{I}_{[i]} \in \mathsf{B}} \Delta_i \\ &\le L_r \sum_{\{j:\mathsf{n}_j > 0\}} \frac{\mathsf{n}_j}{L} \cdot \rho_{[j]} \cdot \Delta_{[j]}^{\mathrm{eq}}, \end{aligned} \tag{57}$$

where the last inequality follows from Eq. (54).

Third, conditioned on $\mathcal{H}_t$, the zero-mean and uncorrelated-noise conditions in Assumption 4 imply:

$$\mathbb{E}\left[\|\mathbf{e}_{\mathsf{noise}}\|_2^2\middle|\mathcal{H}_t\right] = \frac{1}{L^2}\mathbb{E}\left[\left\|\sum_{\mathbf{I}_{[i]}\in\mathsf{B}}\boldsymbol{\xi}_{[i]}\right\|_2^2\middle|\mathcal{H}_t\right] = \frac{1}{L^2}\sum_{\mathbf{I}_{[i]}\in\mathsf{B}}\mathbb{E}\left[\left\|\boldsymbol{\xi}_{[i]}\right\|_2^2\middle|\mathcal{H}_t\right] \le \frac{\sigma^2}{L},$$

which by Jensen's inequality result in:

$$\mathbb{E}\left[\|\mathbf{e}_{\mathsf{noise}}\|_2|\mathcal{H}_t\right] \le \sqrt{\mathbb{E}\left[\|\mathbf{e}_{\mathsf{noise}}\|_2^2\middle|\mathcal{H}_t\right]} \le \frac{\sigma}{\sqrt{L}}. \tag{58}$$

Finally, applying the triangle inequality to Eq. (55), taking the conditional expectation given $\mathcal{H}_t$, and using Eqs. (56) to (58) yield the result in Eq. (20), completing the proof.

F.3 Corollary of Theorem 2

Equation (20) shows that the temporal-error contribution of cluster j is characterized by $\rho_{[j]}\cdot\Delta^{\mathrm{eq}}_{[j]}$, where $\Delta^{\mathrm{eq}}_{[j]}$ in Eq. (19) jointly depends on the occurrence frequency N_j and the elapsed times of experiences in the cluster. However, increasing N_j alone does not necessarily reduce the error, as the elapsed time of a newly added experience affects both $\Delta^{\mathrm{eq}}_{[j]}$ and $\rho_{[j]}$. We provide the following corollary to characterize when increasing occurrence frequency reduces the gradient-error upper bound.

Corollary 2. *Consider increasing the occurrence frequency of cluster j by adding one new experience with elapsed time $\Delta_{[j,\mathsf{N}_j+1]} > 0$, while keeping the existing experiences unchanged. Let $\rho'_{[j]}$ and $\Delta^{\mathrm{eq}\prime}_{[j]}$ denote the corresponding quantities after adding the new experience. The equivalent elapsed time of cluster j decreases if and only if:*

$$\Delta_{[j,\mathsf{N}_j+1]} < \Delta^{\mathrm{eq}}_{[j]}, \tag{59}$$

i.e., the newly added experience is more recent than the equivalent elapsed time of the existing cluster. Furthermore, a sufficient condition for reducing the temporal-error factor of cluster j is

$$\min_{1\le k\le \mathsf{N}_j}\Delta_{[j,k]} \le \Delta_{[j,\mathsf{N}_j+1]} < \Delta^{\mathrm{eq}}_{[j]}, \tag{60}$$

under which $\rho'_{[j]} = \rho_{[j]}$ while $\Delta^{\mathrm{eq}\prime}_{[j]} < \Delta^{\mathrm{eq}}_{[j]}$, yielding:

$$\rho'_{[j]}\cdot\Delta^{\mathrm{eq}\prime}_{[j]} < \rho_{[j]}\cdot\Delta^{\mathrm{eq}}_{[j]}. \tag{61}$$

Thus, for fixed n_j/L, Eq. (61) reduces cluster j's temporal contribution to the error bound.

Remark 5. *Corollary 2 shows that occurrence frequency should not be interpreted independently of recency: adding more experiences is beneficial only when their temporal characteristics improve the corresponding error factor. In particular, sufficiently recent experiences can reduce the equivalent elapsed time without increasing the temporal spread. This coupling provides theoretical support for jointly incorporating frequency and recency into cognitive batch sampling.*

Proof. For notational simplicity, we let:

$$N \triangleq \mathsf{N}_j,\ Z \triangleq \sum_{k=1}^{N}\Delta_{[j,k]}^{-\lambda},\ e \triangleq \left(\frac{N}{Z}\right)^{1/\lambda},\ m \triangleq \min_{1\le k\le N}\Delta_{[j,k]},\ M \triangleq \max_{1\le k\le N}\Delta_{[j,k]},\ \rho \triangleq \frac{M}{m}. \tag{62}$$

We let $d>0$ denote the elapsed time of a newly added experience in cluster j at the same evaluation time. After adding this experience, the corresponding quantities become:

$$N' \triangleq N+1, \qquad Z' \triangleq Z + d^{-\lambda}, \qquad e' \triangleq \left(\frac{N+1}{Z+d^{-\lambda}}\right)^{1/\lambda}, \qquad \rho' \triangleq \frac{\max\{M,d\}}{\min\{m,d\}}. \tag{63}$$

By Eqs. (62) and (63), since $\lambda > 0$ and all elapsed times are positive, $e' < e$ holds if and only if $\frac{N+1}{Z+d^{-\lambda}} < \frac{N}{Z}$. This condition is equivalent to $Z \cdot (N+1) < N \cdot (Z+d^{-\lambda})$, and hence to $Z < N \cdot d^{-\lambda}$. Accordingly, we have:

$$d < \left(\frac{N}{Z}\right)^{1/\lambda} = e = \Delta_{[j]}^{\text{eq}}, \tag{64}$$

where the last equality follows from the definition of $\Delta_{[j]}^{\text{eq}}$ in Eq. (19) and the definitions of N, Z, and e in Eq. (62). Thus, Eq. (64) establishes that the equivalent elapsed time decreases if and only if $d < e$, which is precisely the condition in Eq. (59).

We now consider the complete temporal-error factor associated with cluster j. Before and after adding the new experience, this factor is respectively $\rho \cdot e$ and $\rho' \cdot e'$. Hence, it decreases if and only if $\rho' \cdot e' < \rho \cdot e$.

We next establish the sufficient condition in Eq. (60). Since $m \leq \Delta_{[j,k]} \leq M, k = 1, \ldots, N$, and $\lambda > 0$, we have:

$$M^{-\lambda} \leq \Delta_{[j,k]}^{-\lambda} \leq m^{-\lambda},$$

summing which over all N experiences yields $N \cdot M^{-\lambda} \leq Z \leq N \cdot m^{-\lambda}$, which together with $e = (N/Z)^{1/\lambda}$, gives:

$$m \leq e \leq M. \tag{65}$$

In addition to $d < e$, which guarantees $e' < e$, we further require $d \geq m$. This yields:

$$m \leq d < e. \tag{66}$$

Combining Eq. (66) with Eq. (65), we have:

$$m \leq d < e \leq M,$$

which further implies $\min\{m, d\} = m$ and $\max\{M, d\} = M$. Here, we can conclude that adding the new experience changes neither the minimum nor the maximum elapsed time of cluster j, and consequently,

$$\rho' = \frac{\max\{M, d\}}{\min\{m, d\}} = \frac{M}{m} = \rho. \tag{67}$$

Meanwhile, $d < e$ implies $e' < e$ by Eq. (64). We thus have $\rho' \cdot e' = \rho \cdot e' < \rho e$, which establishes Eq. (61) and proves that Eq. (60) is sufficient. □

G PROOF OF THEOREM 3

The proof proceeds in five steps.

G.1 STEP 1: IMPLICATION OF THE INEQUALITY IN EQ. (23c)

By applying the Schur complement in Lemma 1, the inequality in Eq. (23c) is equivalent to:

$$\alpha \cdot \mathbf{Q}(\mathbf{s}(k)) - [\mathbf{Q}(\mathbf{s}(k)) \cdot \mathbf{A}^\top(\mathbf{s}(k)) + \mathbf{R}^\top(\mathbf{s}(k)) \cdot \mathbf{B}^\top(\mathbf{s}(k))] \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)) \cdot [\mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{Q}(\mathbf{s}(k)) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{R}(\mathbf{s}(k))] \succ 0.$$

Pre- and post-multiplying the above inequality by $\mathbf{Q}^{-1}(\mathbf{s}(k))$ yields:

$$\alpha \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)) - [\mathbf{A}^\top(\mathbf{s}(k)) + \mathbf{Q}^{-1}(\mathbf{s}(k)) \cdot \mathbf{R}^\top(\mathbf{s}(k)) \cdot \mathbf{B}^\top(\mathbf{s}(k))] \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)) \cdot [\mathbf{A}(\mathbf{s}(k)) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{R}(\mathbf{s}(k)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k))] \succ 0,$$

substituting $\mathbf{Q}^{-1}(\mathbf{s}(k)) \triangleq \mathbf{M}(\mathbf{s}(k))$ and $\mathbf{F}(\mathbf{s}(k)) \triangleq \mathbf{R}(\mathbf{s}(k)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k))$ into which gives:

$$\alpha \cdot \mathbf{M}(\mathbf{s}(k)) - [\mathbf{A}^\top(\mathbf{s}(k)) + \mathbf{F}^\top(\mathbf{s}(k)) \cdot \mathbf{B}^\top(\mathbf{s}(k))] \cdot \mathbf{M}(\mathbf{s}(k)) \cdot [\mathbf{A}(\mathbf{s}(k)) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{F}(\mathbf{s}(k))] \succ 0,$$

which is equivalent to

$$\alpha \cdot \mathbf{M}(\mathbf{s}(k)) - \overline{\mathbf{A}}^\top(\mathbf{s}(k)) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \overline{\mathbf{A}}(\mathbf{s}(k)) \succ 0, \tag{68}$$

where $\overline{\mathbf{A}}(\mathbf{s}(k))$ is defined in Eq. (26).

G.2 STEP 2: IMPLICATION OF THE INEQUALITY IN EQ. (23b)

Because $\mathbf{s}(k)$ may contain zero-valued entries, we introduce a perturbed vector $\overline{\mathbf{s}}(k, \delta)$ whose i-th entry is defined as:

$$[\overline{\mathbf{s}}(k, \delta)]_i \triangleq \begin{cases} [\overline{\mathbf{s}}(k)]_i + \delta, & \text{if } [\overline{\mathbf{s}}(k)]_i = 0 \\ [\mathbf{s}(k)]_i, & \text{otherwise} \end{cases}, \quad i = 1, 2, \ldots, n,$$

where $\delta \neq 0$. By construction, the $\overline{\mathbf{s}}(k, \delta)$ contains no zero-valued entries and satisfies:

$$\lim_{\delta \to 0} \overline{\mathbf{s}}(k, \delta) = \mathbf{s}(k). \tag{69}$$

Applying the inequality in Eq. (23b) to $\overline{\mathbf{s}}(k, \delta)$ gives:

$$\lim_{\delta \to 0} \{\mathbf{Q}(\mathbf{s}(k)) - n \cdot \mathsf{diag}^2(\overline{\mathbf{s}}(k, \delta))\} = \mathbf{Q}(\mathbf{s}(k)) - n \cdot \mathsf{diag}^2(\mathbf{s}(k)) \succ 0. \tag{70}$$

The $\lim_{\delta \to 0}\{\mathbf{Q}(\mathbf{s}(k)) - n \cdot \mathsf{diag}^2(\overline{\mathbf{s}}(k, \delta))\} \succ 0$ is equivalent to $\mathbf{Q}(\mathbf{s}(k)) - \lim_{\delta \to 0}\{n \cdot \mathsf{diag}^2(\overline{\mathbf{s}}(k, \delta))\} \succ 0$. Since $\overline{\mathbf{s}}(k, \delta)$ has no zero-valued entries and $\mathbf{Q}(\mathbf{s}(k)) \succ 0$, applying Lemma 4 gives $\lim_{\delta \to 0}\{n^{-1} \cdot \mathsf{diag}^{-2}(\overline{\mathbf{s}}(k, \delta))\} - \mathbf{Q}^{-1}(\mathbf{s}(k)) \succ 0$. Pre- and post-multiplying this inequality by $\mathsf{diag}(\overline{\mathbf{s}}(k, \delta))$ yields:

$$n^{-1} \cdot \mathbf{I}_n - \lim_{\delta \to 0} \{\mathsf{diag}(\overline{\mathbf{s}}(k, \delta)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)) \cdot \mathsf{diag}(\overline{\mathbf{s}}(k, \delta))\} \succ 0. \tag{71}$$

Pre- and post-multiplying Eq. (71) by $\mathbf{1}_n^\top$ and $\mathbf{1}_n$, respectively, and using $\mathbf{Q}^{-1}(\mathbf{s}(k)) = \mathbf{M}(\mathbf{s}(k))$ and $\mathsf{diag}(\overline{\mathbf{s}}(k, \delta)) \cdot \mathbf{1}_n = \overline{\mathbf{s}}(k, \delta)$, we obtain:

$$\lim_{\delta \to 0} \{\mathbf{1}_n^\top \cdot \mathsf{diag}(\overline{\mathbf{s}}(k, \delta)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k)) \cdot \mathsf{diag}(\overline{\mathbf{s}}(k, \delta)) \cdot \mathbf{1}_n = \overline{\mathbf{s}}^\top(k, \delta) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \overline{\mathbf{s}}(k, \delta)\} < n^{-1} \cdot \mathbf{1}_n^\top \cdot \mathbf{I}_n \cdot \mathbf{1}_n = 1,$$

for which, taking $\delta \to 0$ and using Eq. (69) gives:

$$\mathbf{s}^\top(k) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k) = \lim_{\delta \to 0} \{\overline{\mathbf{s}}^\top(k, \delta) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \overline{\mathbf{s}}(k, \delta)\} < 1. \tag{72}$$

G.3 STEP 3: IMPLICATION OF THE INEQUALITY IN EQ. (23a)

We define the following auxiliary set:

$$\mathbb{U}(\mathbf{s}(k)) \triangleq \left\{\widehat{\mathbf{s}} \in \mathbb{R}^n \;\middle|\; \widehat{\mathbf{s}}^\top \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \widehat{\mathbf{s}} \leq 1\right\}. \tag{73}$$

Considering the matrix $\mathbf{C}$ defined in Eq. (2), it follows from Lemma 2 and Eq. (73) that the condition in Eq. (23a) implies:

$$\mathbb{U}(\mathbf{s}(k)) \subseteq \mathbb{S}, \qquad \forall \mathbf{s}(k) \in \mathbb{S}.$$

Therefore, we have:

$$\bigcup_{\mathbf{s}\in\mathbb{S}} \mathbb{U}(\mathbf{s}) \subseteq \mathbb{S}. \tag{74}$$

Moreover, the result in Eq. (72), which follows from Eq. (23b), guarantees that $\mathbf{s} \in \mathbb{U}(\mathbf{s})$ for every $\mathbf{s} \in \mathbb{S}$. Hence,

$$\mathbb{S} \subseteq \bigcup_{\mathbf{s}\in\mathbb{S}} \mathbb{U}(\mathbf{s}). \tag{75}$$

Combining the two inclusions in Eqs. (74) and (75) yields:

$$\bigcup_{\mathbf{s}\in\mathbb{S}} \mathbb{U}(\mathbf{s}) = \mathbb{S}, \tag{76}$$

which is equivalent to $\bigcup_{\mathbf{s}\in\mathbb{S}} \left\{\widehat{\mathbf{s}} \in \mathbb{R}^n \;\middle|\; \widehat{\mathbf{s}}^\top \cdot \mathbf{M}(\mathbf{s}) \cdot \widehat{\mathbf{s}} \leq 1\right\} = \mathbb{S}$ in Eq. (6).

G.4 STEP 4: COMPENSATION ACTION IN EQ. (24)

Substituting the Safety-Teacher action in Eq. (21) into Eq. (31) yields:

$$\begin{aligned}
&\mathbf{s}_{\mathsf{safety}}(k+1) \\
&= \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot (\mathbf{a}_{\mathsf{model}}(k) + \mathbf{a}_{\mathsf{comp}}(k)) + \mathbf{h}(\mathbf{s}(k)) \\
&= \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\mathsf{model}}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\mathsf{comp}}(k) + \mathbf{h}(\mathbf{s}(k)) \\
&= \mathbf{A}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\mathsf{model}}(k) + \mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\mathsf{comp}}(k) + \mathbf{h}(\mathbf{s}(k-1)) + \Delta\mathbf{h}(k),
\end{aligned} \tag{77}$$

where $\Delta\mathbf{h}(k)$ is defined in Eq. (26)

Ideally, the compensation action $\mathbf{a}_{\mathsf{comp}}(k)$ is designed to cancel the model mismatch, which requires:

$$\mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\mathsf{comp}}(k) = -\mathbf{h}(\mathbf{s}(k)). \tag{78}$$

But the model mismatch $\mathbf{h}(\mathbf{s}(k))$ at current time k is unknown, while its value at the previous time can be exactly reconstructed as:

$$\mathbf{h}(\mathbf{s}(k-1)) = \mathbf{s}(k) - \mathbf{A}(\mathbf{s}(k-1)) \cdot \mathbf{s}(k-1) - \mathbf{B}(\mathbf{s}(k-1)) \cdot \mathbf{a}(k-1). \tag{79}$$

We propose to estimate $\mathbf{h}(\mathbf{s}(k))$ via $\mathbf{h}(\mathbf{s}(k-1))$. Replacing $\mathbf{h}(\mathbf{s}(k))$ in Eq. (78) with $\mathbf{h}(\mathbf{s}(k-1))$ results in:

$$\mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\mathsf{comp}}(k) = -\mathbf{h}(\mathbf{s}(k-1)), \tag{80}$$

which, in light of Lemma 5, results in the implementable compensation action in Eq. (24).

G.5 STEP 5: FINAL CONCLUSION

Using Eq. (24) and the range condition in Theorem 3, i.e., $\mathbf{h}(\mathbf{s}(k-1)) \in \mathsf{range}(\mathbf{B}(\mathbf{s}(k)))$, we apply Lemma 5 to obtain:

$$\mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{a}_{\mathrm{comp}}(k) = -\mathbf{B}(\mathbf{s}(k)) \cdot \mathbf{B}^{\dagger}(\mathbf{s}(k)) \cdot \mathbf{h}(\mathbf{s}(k-1)) = -\mathbf{h}(\mathbf{s}(k-1)). \tag{81}$$

Substituting Eqs. (26) and (81), with $\mathbf{Q}^{-1}(\mathbf{s}(k)) = \mathbf{M}(\mathbf{s}(k))$ and $\mathbf{F}(\mathbf{s}(k)) = \mathbf{R}(\mathbf{s}(k)) \cdot \mathbf{Q}^{-1}(\mathbf{s}(k))$, into Eq. (77) yields:

$$\mathbf{s}_{\mathsf{safety}}(k+1) = \overline{\mathbf{A}}(\mathbf{s}(k)) \cdot \mathbf{s}(k) + \Delta\mathbf{h}(k). \tag{82}$$

By Eq. (68), we have:

$$\overline{\mathbf{A}}^{\top}(\mathbf{s}(k)) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \overline{\mathbf{A}}(\mathbf{s}(k)) \prec \widehat{\alpha} \cdot \mathbf{M}(\mathbf{s}(k)). \tag{83}$$

Therefore,

$$\mathbf{s}^{\top}(k) \cdot \overline{\mathbf{A}}^{\top}(\mathbf{s}(k)) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \overline{\mathbf{A}}(\mathbf{s}(k)) \cdot \mathbf{s}(k) < \widehat{\alpha} \cdot \mathbf{s}^{\top}(k) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k). \tag{84}$$

Substituting Eq. (82) into the quadratic form gives:

$$\begin{aligned}\mathbf{s}_{\mathsf{safety}}^{\top}(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\mathsf{safety}}(k+1) = \mathbf{s}^{\top}(k) \cdot \overline{\mathbf{A}}^{\top}(\mathbf{s}(k)) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \overline{\mathbf{A}}(\mathbf{s}(k)) \cdot \mathbf{s}(k) \\ + 2\mathbf{s}^{\top}(k) \cdot \overline{\mathbf{A}}^{\top}(\mathbf{s}(k)) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \Delta\mathbf{h}(k) \\ + \Delta\mathbf{h}^{\top}(k) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \Delta\mathbf{h}(k).\end{aligned} \tag{85}$$

We note from Eq. (72) that, for any $\mathbf{s}(k) \in \mathbb{S}$, we have $\mathbf{s}(k) \in \mathbb{U}(\mathbf{s}(k))$, with

$$\mathbf{s}^{\top}(k) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}(k) < 1. \tag{86}$$

Combining Eqs. (84) to (86) gives:

$$\begin{aligned}\mathbf{s}_{\mathsf{safety}}^{\top}(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\mathsf{safety}}(k+1) < \widehat{\alpha} + 2\mathbf{s}^{\top}(k) \cdot \overline{\mathbf{A}}^{\top}(\mathbf{s}(k)) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \Delta\mathbf{h}(k) \\ + \Delta\mathbf{h}^{\top}(k) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \Delta\mathbf{h}(k).\end{aligned} \tag{87}$$

Since $\mathbf{M}(\mathbf{s}(k)) = \mathbf{Q}^{-1}(\mathbf{s}(k))$, the condition in Eq. (25) implies that the right-hand side of Eq. (87) is strictly smaller than α. Therefore,

$$\mathbf{s}_{\mathsf{safety}}^{\top}(k+1) \cdot \mathbf{M}(\mathbf{s}(k)) \cdot \mathbf{s}_{\mathsf{safety}}(k+1) < \alpha, \qquad \forall\, \mathbf{s}(k) \in \mathbb{S}. \tag{88}$$

Hence, under Assumption 1, we have:

$$\mathbf{f}^{\top}(\mathbf{s}, \mathbf{a}_{\mathsf{safety}}) \cdot \mathbf{M}(\mathbf{s}) \cdot \mathbf{f}(\mathbf{s}, \mathbf{a}_{\mathsf{safety}}) < \alpha, \qquad \forall\, \mathbf{s} \in \mathbb{S}, \tag{89}$$

which establishes the first condition in Eq. (6). Together with the set-coverage condition established in Eq. (72), the Safety-Teacher satisfies the design requirement in Eq. (6). This completes the proof.

H OPEN-WORLD EXPERIMENT DESIGN: REAL WILD FORESTS

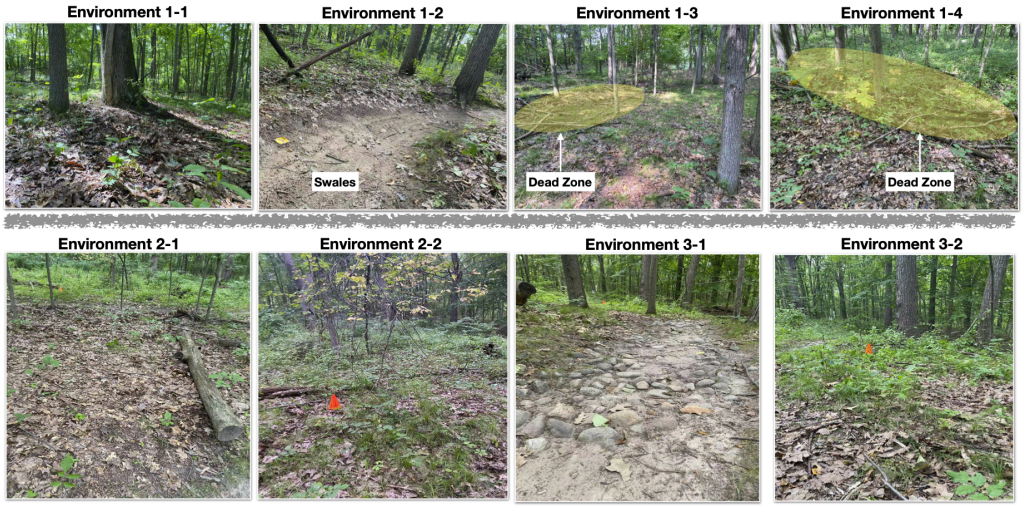


Figure 6: Open-World Environments: Wild Forests.

We perform **OpenRun** on edge-AI device, i.e., NVIDIA Jetson Orin, which thus enables the quadruped robot to perform runtime learning for end-to-end vision-to-action navigation in challenging wild forest environments without dependence on remote cloud and GPS. The navigation goal is set approximately 30 meters from the home base. These open-world environments are shown in Fig. 6.

Open-world environments also imply that operational scenarios cannot be fully known or represented prior to deployment. To account for such uncertainty, we define safety conditions conservatively based on worst-case operational requirements. Accordingly, the safety set is defined as:

$$\mathbb{S} = \{\mathbf{s} \mid \text{|obstacle distance|} \geq 0.1 \text{ m}, \ \text{|pitch rate|} \leq 45^\circ/\text{sec}, \ \text{ and |roll rate|} \leq 45^\circ/\text{sec}\}, \quad (90)$$

whose safety conditions are to avoid collisions with obstacles and avoid balance loss. We next present the details of this experiment design.

H.1 LEARNING-STUDENT MODULE

The design of the Learning-Student module encompasses the reward function, the specifications of actor and critic networks, and cognitive batch sampling.

H.1.1 REWARD

The reward functions for the Learning-Student primarily focus on the following key aspects:

- **Body Stability:** We adopt a reward function that accounts for both stability and partial safety considerations:

$$\mathcal{R}_{\text{stability}} = -50(p_{\text{height}} - 0.3)^2 - 5p_{\text{roll}}^2 - 5p_{\text{pitch}}^2 - 2v_{\text{height}}^2 - 2v_{\text{roll}}^2 - 2v_{\text{pitch}}^2, \quad (91)$$

  where p_{height}, p_{roll}, and p_{pitch} denote the height, roll, and pitch, respectively, while v_{height}, v_{roll}, and v_{pitch} denote their corresponding rates.

- **Path Safety:** We propose a reward function that encourages the robot to seek safer paths by maximizing the minimum distance from surrounding obstacles:

$$\mathcal{R}_{\text{safety}} = -\frac{1}{d_{\min}}, \quad (92)$$

  where $d_{\min}$ denotes the minimum distance between the robot and its surrounding obstacles.

- **Path Length:** We include a reward function that encourages the robot to seek shorter paths toward the goal:

$$\mathcal{R}_{\mathsf{length}} = -||p_{\text{position}} - p_{\text{goal}}||_2^2, \tag{93}$$

where p_{position} and p_{goal} denote the robot's current position and the preset goal position, respectively.

- **Lateral Momentum:** We include a reward function to minimize travel time to the goal and reduce lateral momentum by penalizing lateral velocity and encouraging forward velocity:

$$\mathcal{R}_{\mathsf{lateral}} = -\left(\omega_1 \cdot \frac{1}{|v_x| + \epsilon} + \omega_2 \cdot v_y^2\right), . \tag{94}$$

where v_x and v_y are the forward and lateral CoM velocities, respectively, ω_1 and ω_2 are weighting coefficients.

- **Action Jerk:** To discourage abrupt actions, we define the following reward function penalizing action jerk:

$$\mathcal{R}_{\mathsf{jerk}} = -||\mathbf{a}||^2. \tag{95}$$

The composite reward function for guiding the Learning-Agent in acquiring a safe and high-performance navigation policy through runtime learning is formulated by combining the aforementioned reward components Eqs. (91) to (95):

$$\mathcal{R} = \xi_1 \cdot \mathcal{R}_{\mathsf{stability}} + \xi_2 \cdot \mathcal{R}_{\mathsf{safety}} + \xi_3 \cdot \mathcal{R}_{\mathsf{length}} + \xi_4 \cdot \mathcal{R}_{\mathsf{lateral}} + \xi_5 \cdot \mathcal{R}_{\mathsf{jerk}},$$

where the weights are set to $\xi_1 = 0.1, \xi_2 = 0.1, \xi_3 = 0.01, \xi_4 = 0.002$, and $\xi_5 = 0.0001$.

H.1.2 Actor and Critic Specifications

The observation input to the actor network is defined as:

$$\mathbf{o} = [\mathbf{v}_{\mathsf{feature}}, \text{CoM x-position}, \text{CoM y-position}, \text{CoM z-position}, \text{roll}, \text{pitch}, \text{yaw}, \text{CoM x-velocity}, \text{CoM y-velocity}, \text{CoM z-velocity}, \text{roll rate}, \text{pitch rate}, \text{yaw rate}] \in \mathbb{R}^{36},$$

where $\mathbf{v}_{\mathsf{feature}} \in \mathbb{R}^{24}$ is the extracted visual feature vector. The output of actor network is a 6-DoF acceleration command $\mathbf{a} = [\dot{\mathbf{v}}, \dot{\boldsymbol{\omega}}]$. The action space of Learning-Student is summarized as $\mathbf{a}_{\mathsf{learning}} \in [-5, 10] \times [-2.5, 2.5] \times [0, 10] \times [-15, 15]^3$.

The actor and critic networks are each realized as MLPs with four fully connected layers. The actor network has layer output dimensions of 256, 256, 256, and 6, while those of the critic network are 256, 256, 256, and 1. Both networks employ ReLU activations in the first three layers; the actor's output layer applies a Tanh activation, while the critic's output layer applies a linear activation. The discount factor is set to 0.95, with learning rates of 1e-5 and 5e-7 for the critic and actor networks, respectively, and a batch size of 256.

H.1.3 Cognitive Sampling Specifications

For the frequency and recency measures in Eq. (17), the memory decay factor, cluster size, and cluster criterion are set to $\lambda = 0.5$, $C = 50$, and $\mathsf{c} = 0.2$, respectively. While for the similarity measure, we consider the simple and computationally efficient one, which is the inverse Euclidean-distance similarity, i.e.,

$$\mathcal{S}(\mathbf{I}_{[i]}, \mathbf{S}_t) = \xi \cdot \left(1 + \left\|\mathbf{I}_{[i]} - \mathbf{S}_t\right\|_2\right)^{-1}, \tag{96}$$

where $\|\cdot\|_2$ denotes the Euclidean norm, measuring the distance between the prior experience instance $\mathbf{I}_{[i]}$ and the real-time experience instance $\mathbf{S}_t$ acquired at time $t \in \mathbb{N}$, and $0 < \xi < 1$ is a hyperparameter controlling the relative importance of similarity in the overall ranking in Eq. (11). Here, we let $\xi = 0.3$.

H.2 Safety-Teacher Module

The only design parameter of Safety-Teacher is the parameter of $\alpha = 0.55$ and $\widehat{\alpha} = 0.5$ in Theorem 3.

H.3 EDGE-AI DEVICE SPECIFICATIONS

Open-world operation often involves infrastructure-limited environments, such as dense forests, disaster-affected areas, and deep-space environments, where reliable access to external infrastructure cannot be guaranteed. GPS signals may be degraded or entirely unavailable, wireless communication may be intermittent or absent, and remote cloud computing resources may be inaccessible. Consequently, autonomous systems cannot consistently rely on external localization, communication, or cloud-based computation. To address this challenge, we deploy **OpenRun** entirely on an onboard edge-AI device. In this work, **OpenRun** runs on an NVIDIA Jetson AGX Orin, whose key specifications are summarized as follows:

- **AI Performance:** Up to 275 TOPS
- **CPU:** 12-core Arm Cortex-A78AE (v8.2, 64-bit, up to 2.2 GHz)
- **GPU:** NVIDIA Ampere GPU (2048 CUDA cores, 64 Tensor Cores)
- **Memory:** 64 GB LPDDR5 (256-bit, 204.8 GB/s bandwidth)
- **Storage:** 1 TB WD_BLACK SN850X NVMe SSD
- **Power Configuration:** Configurable between 15 W and 60 W
- **Software Stack:** JetPack 6.2 (L4T Ubuntu 22.04) with ROS 2 Humble

H.4 OPERATION PRACTICES ON EDGE

Another motivation behind operating **OpenRun** on edge-AI devices is that edge-AI devices support lightweight, locally executable models, enabling direct on-device and real-time interaction with the open world, which preserves temporal consistency and low latency. In this work, we adopt a representative edge-AI device, the **NVIDIA Jetson**, which comprises an ARM-based CPU and a GPU. We outline the key operational practices below.

Efficiency and Scalability: OpenRun deploys its Safety-Teacher on an ARM-based CPU, leveraging its superior performance-per-watt ratio and thermal efficiency, both of which are critical for mobile autonomous systems and long-duration on-device runtime learning. This design enables energy-efficient execution and scalable, robust decision-making for safety-critical edge deployment.

Decoupled Multi-Rate Execution: OpenRun leverages the multi-rate architecture of edge-AI devices. Specifically, the Action-Blender and Safety-Teacher are implemented in C/C++ on the CPU to ensure high-frequency operation (200–1000 Hz) for safe control, while the Learning-Student runs at lower frequencies (10–20 Hz) on the GPU. By decoupling learning from safe control, the system ensures stable, safe interaction with time-critical physical environments while preserving real-time execution. This design further improves runtime learning stability without compromising real-time system safety.

I Open-World Experiment Design: Simulated Wild Forests

I.1 Workstation Specifications

For experiments in the simulated wild forests, **OpenRun** runs on a workstation with the following specifications:

- **AI Performance:** Up to 4000 TOPS
- **CPU:** 24-core Intel Core Ultra 9 285K (up to 5.1 GHz)
- **GPU:** NVIDIA RTX PRO 6000 Blackwell GPU with 96 GB GDDR7 VRAM and fifth-generation Tensor Cores
- **Memory:** 64 GB DDR5-4800 RAM (4×16 GB DIMMs)
- **Storage:** 2 TB Samsung 990 PRO NVMe SSD
- **Software Stack:** Ubuntu 22.04 with ROS 2 Humble and CUDA 13.1

I.2 Off-Road Autonomous Vehicles: Navigation Planning

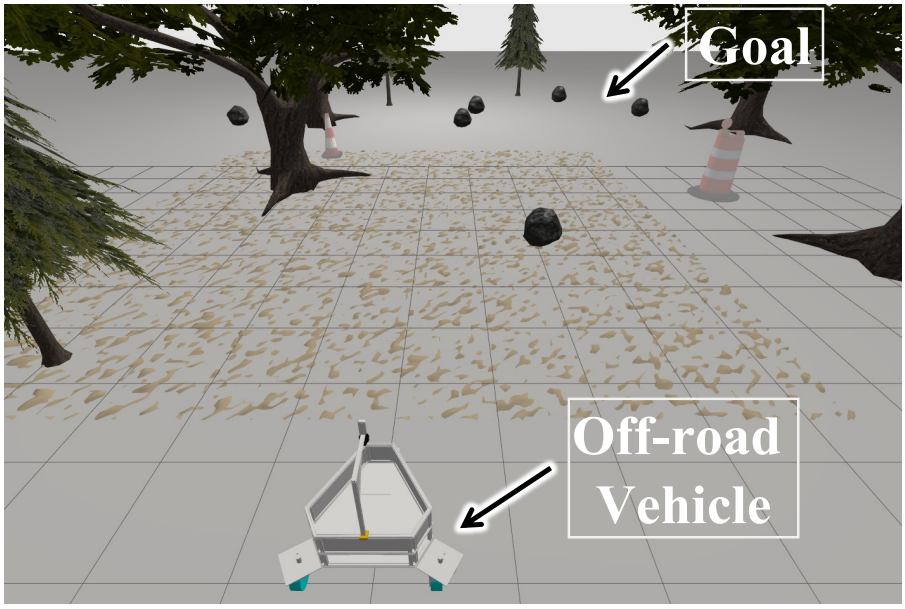


Figure 7: Off-road autonomous vehicle in the open-world environments.

In this experiment, **OpenRun** enables the off-road autonomous vehicle to perform runtime learning for navigation planning. As shown in Fig. 7, the unstructured environment features randomized obstacles, including rocks and trees, across terrains ranging from flat to uneven surfaces.

The safety set is defined as:

$$\mathbb{S} = \{\mathbf{s} \mid |\text{obstacle distance}| \geq 0.55 \text{ m}, \ \ |\text{pitch rate}| \leq 1 \text{ rad/sec}, \ \ |\text{roll rate}| \leq 1 \text{ rad/sec}\}.$$

I.2.1 Learning-Student Module

The design of the Learning-Student module is outlined below.

Reward: The reward function of the Learning-Agent is formulated as:

$$\mathcal{R} = \xi_1 \cdot \mathcal{R}_{\mathsf{stability}} + \xi_2 \cdot \mathcal{R}_{\mathsf{safety}} + \xi_3 \cdot \mathcal{R}_{\mathsf{length}} + \xi_4 \cdot \mathcal{R}_{\mathsf{lateral}},$$

where $\mathcal{R}_{\mathsf{safety}}$, $\mathcal{R}_{\mathsf{length}}$, and $\mathcal{R}_{\mathsf{lateral}}$ are defined in Eqs. (92) to (94), respectively, and

$$\begin{aligned}\mathcal{R}_{\mathsf{stability}} = &-5p_{\mathsf{roll}}^2 - 5p_{\mathsf{pitch}}^2 - 2v_{\mathsf{height}}^2 - 2v_{\mathsf{roll}}^2 - 2v_{\mathsf{pitch}}^2 - 2(v_{\mathrm{x}} - \bar{v}_{\mathrm{x}})^2 - 2(v_{\mathrm{y}} - \bar{v}_{\mathrm{y}})^2 \\ &- 2(v_{\mathsf{yaw}} - \bar{v}_{\mathsf{yaw}})^2,\end{aligned}$$

where p_{height}, p_{roll}, and p_{pitch} denote the height, roll, and pitch, respectively, while v_{height}, v_{roll}, and v_{pitch} denote their corresponding velocities. The v_{x}, v_{y}, and v_{yaw}, respectively, denote vehicle's CoM-x velocity, CoM-y velocity, and yaw rate. The $\bar{v}_{\text{x}}$, $\bar{v}_{\text{y}}$, and $\bar{v}_{\text{yaw}}$ denote the corresponding planned values from the Learning-Agent or Rational-Agent. This reward function accounts for system stability and safety in tracking the planned motions $(\bar{v}_{\text{x}}, \bar{v}_{\text{y}}, \bar{v}_{\text{yaw}})$. The associated weighting coefficients are $\xi_1 = 0.1, \xi_2 = 0.05, \xi_3 = 0.01$, and $\xi_4 = 0.002$.

Actor and Critic Specifications: The observation input to the actor network is defined as:

$$\mathbf{o} = [\mathbf{v}_{\text{feature}}, \text{CoM x-position}, \text{CoM y-position}, \text{CoM z-position}, \text{roll}, \text{pitch}, \text{yaw}, \text{CoM x-velocity}, \text{CoM y-velocity}, \text{CoM z-velocity}, \text{roll rate}, \text{pitch rate}, \text{yaw rate}] \in \mathbb{R}^{36},$$

where $\mathbf{v}_{\text{feature}} \in \mathbb{R}^{24}$ is the extracted visual feature vector. The actor network outputs the planned motion commands, comprising the desired yaw rate, CoM x-velocity, and CoM y-velocity, which are passed to the low-level controller for tracking. The input to the critic network comprises the observation input $\mathbf{o}$ and the actions.

The actor and critic networks are each realized as MLPs with four fully connected layers. The actor network has layer output dimensions of 256, 256, 256, and 3, while those of the critic network are 256, 256, 256, and 1. Both networks employ ReLU activations in the first three layers; the actor's output layer applies a Tanh activation, while the critic's output layer applies a linear activation. The discount factor γ is set to 0.95, with learning rates of 1e-5 and 1e-6 for the critic and actor networks, respectively, and a batch size of 512.

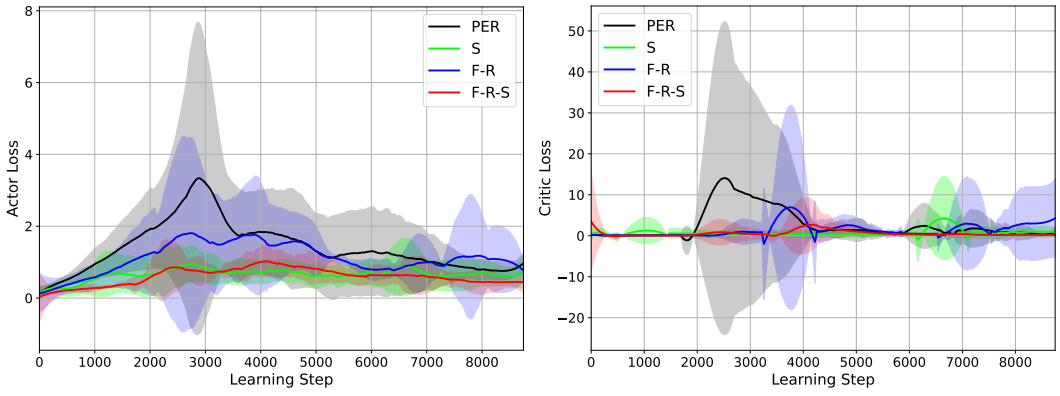
Cognitive Batch Sampling Specifications: For the frequency and recency measures in Eq. (17), the memory decay factor is set to $\lambda = 0.5$, the cluster size to $\mathsf{C} = 50$, and the family criterion to $\mathsf{c} = 0.5$. The similarity measure function adopts the one in Eq. (96).

I.2.2 Safety-Teacher Module

We adopt Interactive-Far (He et al., 2024) only as the Safety-Teacher, which is a navigation planning model that outputs verifiable collision-free motions.

I.2.3 Experimental Result

Fig. 8 shows the actor and critic losses over five random seeds, each with 30 episodes.



(a) Actor Loss (95% CI). (b) Critic Loss (95% CI).

Figure 8: Ablation of cognitive batch sampling and comparison with prioritized experience replay (PER).

I.3 Quadruped Robot: Locomotion

This experiment focuses on evaluating the non-learning design of the **Safety-Teacher**, with the testing environment shown in Fig. 9.



Figure 9: Terrain conditions for controller evaluation via locomotion control.

Current safe RL frameworks typically rely on physics-model-based controllers for safety assurance. Existing approaches can be broadly categorized into two groups: 1) static-model control (denoted as 'SM"), as adopted in residual RL (Johannink et al., 2019a; Cao et al., 2024) and Runtime-Assurance RL frameworks such as MPS (Bastani, 2021; Banerjee et al., 2024) and Neural Simplex (Phan et al., 2020); and 2) adaptive-model control (denoted as "AM"), as recently introduced in Runtime-Assurance RL frameworks such as the Runtime Learning Architecture (Cai et al., 2025) and Real-DRL (Mao et al., 2025). In parallel, the control community has extensively studied fixed-model adaptive control (denoted as "FMAC"). Our proposed adaptive-model adaptive controller (denoted as "AMAC") goes beyond these approaches by jointly adapting the system model and compensating for residual model mismatch, thereby improving the ability to maintain stable operation under challenging open-world conditions.

For evaluation, we task the robot's model-based controller with locomotion over highly uneven and challenging terrain, as shown in Fig. 9. Each approach is evaluated using five random seeds. Control performance is quantified using the stability metric defined in Eq. (91), with the corresponding trajectories shown in Fig. 10. The results show that the proposed "AMAC" consistently achieves better stability than "SM", "AM", and "FMAC" when traversing challenging terrain, demonstrating the benefit of jointly adapting the system model and compensating for model mismatch.

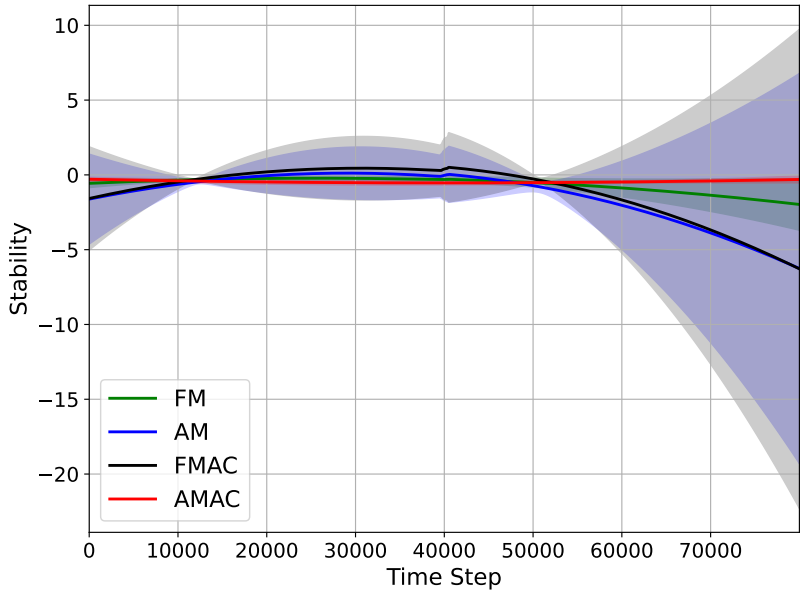


Figure 10: Stability performance.